

Reconstructing the Commercial Landings of Red Snapper in the Gulf of Mexico from 1872 to 1963

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Abstract.—Statistics on the commercial landings of red snapper *Lutjanus campechanus* in the United States have been recorded by the National Marine Fisheries Service (NMFS) and its predecessors as far back as 1880. However, there are many gaps in the historical record. Censuses were conducted sporadically prior to 1949 and the water body where the catch was taken was not consistently recorded until 1963. Past assessments of the status of red snapper in the U.S. Gulf of Mexico have avoided these problems by using only data collected after the early 1960s. Unfortunately, there are indications the stock may have already been depressed by this time and the resulting lack of contrast in the data has made it difficult to estimate important benchmarks such as the maximum sustainable yield. For this reason, the participants of several SEDAR (Southeast Data and Assessment Review) workshops strongly recommended reconstructing the catches as far back as possible using whatever auxiliary information might be available. This paper uses statistics from several state and private agencies to supplement NMFS landings statistics and then uses historical accounts dating back to the 1840s to help fill the remaining gaps. The data suggest that a substantial red snapper fishery existed as early as 1872 and that by the turn of the century the landings were comparable to those of recent times, albeit heavily dependent on snapper grounds located in Mexico. The trends of the reconstructed landings are qualitatively consistent with major events in the history of the fishery, but considerable uncertainty remains over the many uncanvassed years between 1890 and the 1920s.

Introduction

The fishery for red snapper *Lutjanus campechanus* began in the early 1840s with the accidental discovery of a large aggregation by Captain James Keeny, a fisherman

from Connecticut who used to come south each winter to fish the Gulf. Collins (1887) reports Captain Keeny's account of the adventure as follows: "On one occasion when I was on my way to New Orleans with a cargo of beach fish (pompano, sheepshead, redfish,

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etc.), I got becalmed when several miles offshore. We had just finished eating, and the cook came on deck and threw over some refuse from the table. The vessel lay motionless, and very soon many strange looking red fish were seen in the water alongside, eagerly feeding on the material the cook had thrown overboard. We quickly baited some lines and threw them out, and the fish bit as fast as we could haul them in. Nearly two hundred snappers were caught, which we took to New Orleans, where they sold like hot cakes.”

The Gulf of Mexico was poorly charted in those days, and fishermen knew little of the offshore grounds. It is said that Captain Keeny and his crew did not even realize they were “on soundings,” i.e., in less than 100 fathoms, when they made their discovery (Collins 1887). Nevertheless, within a few years several red snapper banks were located and the potential of the resource was quickly recognized. The first serious attempts to develop a commercial market began in the late 1840s, but the trade was mostly local owing to difficulties in transporting and preserving the catch (Warren 1898). The commercial fishery is said to have begun in earnest in 1872 when Mr. S. C. Cobb brought four New England live-well “smacks” down to Pensacola, Florida and built the first fish house dedicated to processing and shipping red snapper (Jarvis 1935; Bortone et al. 1997). Pensacola proved to be an ideal port for landing red snapper because, in addition to having a railroad terminal, the city wharves were in saltwater, making it possible to land the fish alive and fresh (Stearns 1887). Initially the primary markets were in New Orleans and Mobile, but by the late 1870s the market had expanded to include cities throughout the Mississippi valley and as far north as New York (Goode and Gill 1903).

The snapper grounds exploited during the early days of the fishery were located primarily along the edge of the continental shelf between Mobile Bay, Alabama, and Cape St. George, Florida. Those grounds were soon depleted however, and the fleet moved progressively further south towards the Florida Middle Grounds (a 1,000 km² bank about 140 km south of Apalachicola and 120 km northwest of Tarpon Springs). By 1883 the Middle Grounds

were also becoming unproductive, prompting Stearns (1883) to remark “Most of the old fishing grounds are barren, and smacks have to go farther each year to find new ones.” In 1885 the research vessel *Albatross* found productive snapper grounds even further to the south off Tampa and the Dry Tortugas (Collins 1885). At about the same time new snapper grounds were also found off Texas (Camber 1955). These discoveries, in concert with the strong market demand, revitalized the fishery and new fish houses were established throughout the Gulf of Mexico. In the early 1890s a few vessels began making trips to productive snapper grounds found on the Campeche Banks off Mexico, but preserving the catch proved difficult until about the turn of the century when cheap manufactured ice became widely available. After that the landings soared to a record high of almost 14 million pounds in 1902 (Figure 1). By 1910 most snapper vessels were spending at least part of their time fishing off the Campeche Banks or elsewhere in Mexico, usually during the winter when the threat of hurricanes was over (Camber 1955). This pattern continued until the early 1980s when the government of Mexico excluded the U.S. fleet.

Statistics on the historical landings of red snapper in the United States have been collected by the National Marine Fisheries Service (NMFS) and its predecessors since 1880, but there are several gaps in the historical record (Figure 1). Moreover, prior to 1963, the statistical summaries that are typically available include only the port of landing and not the water body where the catch was taken. These omissions are of particular concern inasmuch as a large fraction of the total U.S. landings is known to have been from outside of U.S. waters. For this reason, past assessments of the status of red snapper in the Gulf of Mexico have relied on the more complete data collected since the 1960s. Unfortunately, there are indications the abundance of the stock may have already been depressed by this time and the corresponding data do not exhibit any strong temporal trends. As a result it has been difficult to reliably estimate important benchmarks such as the maximum sustainable yield. To remedy this, the participants of several South-

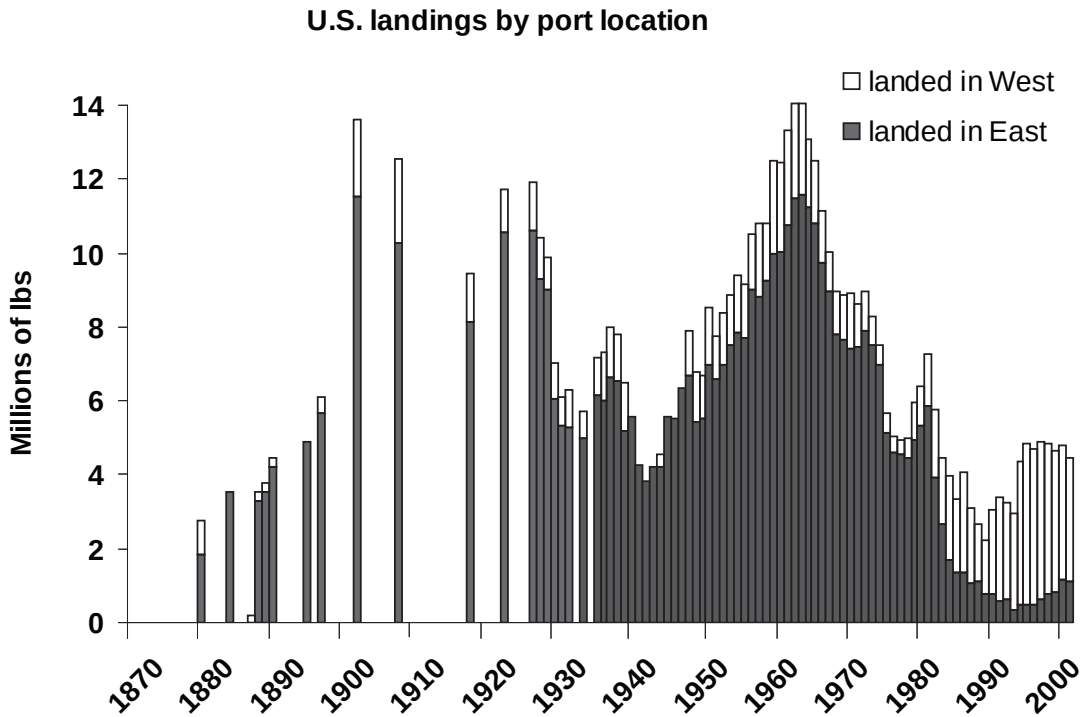


Figure 1. Census estimates of red snapper landings at U.S. ports located east or west of the Mississippi River (without regard to fishing location).

east Data and Assessment Review workshops recommended reconstructing the catches as far back as possible using whatever auxiliary information might be available (SEDAR 2004). This paper details a first attempt at such a reconstruction using clues from several historical references.

Methods

As mentioned previously, the Federal records prior to 1963 are incomplete. Censuses were not always conducted annually in every region (owing to budget limitations, of course). Moreover, the older statistics were published in greater or lesser detail under the auspices of a succession of Federal agencies including the U.S. Fish Commission, Bureau of Fisheries, Fish and Wildlife Service, Bureau of Commercial Fisheries, and Bureau of the Census. The NMFS Fishery Statistics Division has summarized many of these reports (Anonymous 1990), but they did not include information from studies that were focused particularly on the red snapper fishery.

Camber (1955) provided more complete statistics by including data collected during Federal-sponsored studies of the red snapper fishery as well as statistics published since 1938 by the Florida State Board of Conservation. Even so, substantial gaps remain in the historical record and neither set of summaries records where the fish were caught. The following sections detail how these gaps were filled based on auxiliary information gleaned from the literature.

Estimating landings during years that were not surveyed

The landings statistics are especially spotty prior to 1927, being limited to 1880, 1884, 1889, 1890, 1895, 1902, 1908, 1918, and 1923. However, Townsend (1900) gives annual statistics on the number of red snapper vessels operating out of the primary port of Pensacola from 1872 to 1897. This information suggests that, while the number of vessels increased rapidly during the late 1800s, the average landings per vessel was relatively stable at around 120,000 lbs (Table 1). Thus, it seems reasonable to estimate the landings

Table 1. Number of vessels fishing for red snapper out of Pensacola (Camber 1955) and the ratio of total landings (in thousands of pounds) to number of vessels for years when landings surveys were conducted. Gray areas denote linear interpolations for years with no vessel information.

Year	Number of vessels	Landings per vessel
1872	4	
1873	6	
1874	9	
1875	11	
1876	13	
1877	11	
1878	10	
1879	11	
1880	14	105.9
1881	21	
1882	26	
1883	24	
1884	25	142.1
1885	27	
1886	33	
1887	34	
1888	34	
1889	35	99.1
1890	34	122.7
1891	36	
1892	37	
1893	39	
1894	40	
1895	42	116.3
1896	36	
1897	36	147.6

Partitioning landings by fishing location

This section describes how the U.S. red snapper landings were partitioned according to the water body where the fish were caught. Three water bodies are considered: waters outside of U.S. jurisdiction (primarily Mexico) and waters within U.S. jurisdiction lying east or west of the Mississippi River. The division at the Mississippi river accommodated requests by SEDAR 7 panelists that were based on recent evidence in support of possibly distinct populations in the eastern and western Gulf.

in Florida for years that were not surveyed from the number of vessels and the average landings per vessel observed during the nearest surveyed years (as shown in Table 2). Unfortunately, we were unable to locate similar annual indices of effort for time periods after 1897 or for ports other than Pensacola. While Jarvis (1935) and Camber (1955) do provide some information in this regard, their vessel counts come from the same censuses as the landings data and therefore have the same gaps. Moreover, the average efficiency of the vessels increased markedly with the advent of steam and diesel powered motors during the 1920s and again after World War II with the increased availability of diesel engines, depth recorders and other gear innovations. Nevertheless, the technology and number of vessels does not appear to have changed too much from one census to the next, suggesting that the landings during the intervening periods may be reasonably approximated by linear interpolation (Table 2).

Florida and Alabama

1872–1933.—The commercial fishery essentially began in 1872 with four vessels fishing locally off Pensacola. By 1890, the Pensacola fleet had grown to 34 vessels and smaller fleets were established in several other Florida ports as well as in Mobile, Alabama. All of these fleets fished almost exclusively off Florida until late in 1890 when a few vessels began fishing on the Campeche Banks off Mexico (Stearns 1883, 1885, 1887; Smith 1894). The fraction of U.S. landings coming from the Campeche banks appears to have been rather low until about 1895 when ice became more readily available, making it easier to preserve the catch during the long return trip from Mexico. By 1910 most of the larger vessels (over 5 tons) from Pensacola and Mobile were fishing primarily on the Campeche Banks (Camber 1955). Jarvis (1935) and Carpenter (1965) estimated that 75% of all landings by vessels (crafts over 5 tons) were from Mexico

Table 2. Total landings by state from 1872 to 1963 (the inception of the commercial fishery to the point where the U.S. fishery statistics routinely included the water body from which the catch was taken). The shaded regions refer to gaps that were filled as described in the text.

year	State where catch was landed					
	TX	LA	MS	LA	FL	Gulfwide
1872	0	0	0	0	423795	423795
1873	0	0	0	0	671009	671009
1874	0	0	0	0	918223	918223
1875	0	0	0	0	1165438	1165438
1876	0	0	0	0	1377335	1377335
1877	0	0	0	0	1165438	1165438
1878	0	0	0	0	1059489	1059489
1879	0	0	0	0	1165438	1165438
1880	0	900000	0	360000	1483284	2743284
1881	0	810000	0	323000	2710916	3843916
1882	0	719000	0	287000	3356373	4362373
1883	11667	629000	0	250000	3098190	3988857
1884	23333	539000	0	213000	3551275	4326608
1885	35000	448000	0	176000	3159290	3818290
1886	46667	358000	0	140000	3861354	4406021
1887	75000	131000	0	103000	3939362	4248362
1888	65000	150000	0	86000	3224000	3525000
1889	22000	250000	0	51000	3469369	3792369
1890	5000	240000	0	62000	4172942	4479942
1891	93000	183000	0	105000	4243585	4624585
1892	155000	152000	0	143000	4434308	4884308
1893	217000	122000	0	181000	4625030	5145030
1894	279000	91000	0	220000	4815753	5405753
1895	341000	61000	0	258000	4886396	5546396
1896	403000	30000	0	297000	4708100	5438100
1897	465000	0	0	335000	5314487	6114487
1898	786000	0	0	961000	5695000	7442000
1899	1106000	0	0	1587000	6290000	8983000
1900	1427000	0	0	2214000	6885000	10526000
1901	1747000	0	0	2840000	7479000	12066000
1902	2068000	0	0	3466000	8074066	13608066
1903	2099000	0	0	3328000	8005000	13432000
1904	2129000	0	0	3189000	7936000	13254000
1905	2160000	0	0	3051000	7867000	13078000
1906	2191000	0	0	2912000	7797000	12900000
1907	2221000	0	0	2774000	7728000	12723000
1908	2252000	0	0	2635000	7659000	12546000
1909	2151000	6000	10000	2451000	7616000	12234000
1910	2050000	12000	20000	2268000	7573000	11923000
1911	1949000	18000	29000	2084000	7530000	11610000
1912	1848000	24000	39000	1900000	7487000	11298000
1913	1748000	30000	49000	1717000	7445000	10989000
1914	1647000	36000	59000	1533000	7402000	10677000
1915	1546000	42000	69000	1349000	7359000	10365000
1916	1445000	48000	78000	1165000	7316000	10052000
1917	1344000	54000	88000	982000	7273000	9741000
1918	1243000	60000	98000	798000	7230000	9429000
1919	1196000	83000	99000	832000	7678000	9888000
1920	1149000	106000	100000	867000	8126000	10348000
1921	1103000	129000	102000	901000	8575000	10810000
1922	1056000	152000	103000	936000	9023000	11270000
1923	1009000	175000	104000	970000	9471000	11729000
1924	1015000	148000	112000	1026000	9179000	11480000

Table 2. Continued.

1925	1021000	121000	120000	1083000	8886000	11231000
1926	1026000	94000	128000	1139000	8594000	10981000
1927	1237306	72000	218706	1058650	9312667	11899329
1928	1055162	48000	97328	1300522	7891835	10392847
1929	804140	80000	90864	1227601	7700143	9902748
1930	929578	76400	188725	847918	5001672	7044293
1931	690664	78835	68236	863201	4392992	6093928
1932	985291	66884	36812	681573	4539465	6310025
1933	810346	72942	80006	816037	4227833	6007163
1934	635400	79000	123200	950500	3916200	5704300
1935	771000	98000	224050	989000	4360150	6442200
1936	906600	117000	324900	1027500	4804100	7180100
1937	1141200	148100	303800	1168200	4550500	7311800
1938	1279000	85000	173900	1193100	5260800	7991800
1939	1156300	90900	40400	1020000	5493700	7801300
1940	1233100	104400	26200	1255400	3891300	6510400
1941	1223000	93000	80000	1492263	4058484	6946747
1942	849000	66000	53000	1275062	2988486	5231548
1943	662000	53000	39000	1001293	2824923	4580216
1944	475000	39000	26000	1092384	3108629	4741013
1945	288200	25500	12000	1360500	2845800	4532000
1946	500000	46000	26000	1703822	3866392	6142214
1947	712000	66000	40000	1514391	4029488	6361879
1948	923000	87000	54000	1851800	4508038	7423838
1949	1054400	169900	135900	1343200	5184500	7887900
1950	1233200	142000	65600	993900	4353600	6788300
1951	1116800	9200	2400	1228800	4313200	6670400
1952	1523400	64600	0	1458800	5499900	8546700
1953	1100700	44100	28900	1418000	5135600	7727300
1954	1344700	45200	68000	1403600	5524000	8385500
1955	1261700	70900	147000	1173200	6209700	8862500
1956	1533500	43700	271400	1065100	6499716	9413416
1957	1443000	28200	549600	932900	6201459	9155159
1958	1399000	87800	1109900	1417900	6487173	10501773
1959	1665100	313000	1021700	1819100	5993778	10812678
1960	1152600	426000	1468500	1720100	6046170	10813370
1961	1828900	677200	2151700	1784200	6044949	12486949
1962	1742300	693900	2175900	1893400	5966250	12471750
1963	2168700	387900	1885800	2314900	6568758	13326058

until about 1933, when the proportion dropped to about 50%. For lack of more detailed information, we assume that the proportion of the vessel landings that came from U.S. waters decreased linearly from 100% in 1894 to 25% in 1910 and then remained constant at about 25% until 1933, when it increased to 50%. The remaining vessel landings are assumed to have come from the Dry Tortugas grounds (cf. Jarvis 1935).

The landings in Florida and Alabama by boats (defined as crafts under 5 tons) were small early in the fishery, but increased to more substantive levels during the 1920s (Table 3). Owing to their small size, most boats fished fairly close to the ports where the landings

were made rather than in distant foreign waters. Hence the proportion of the total landings for boats and vessels combined that came from U.S. waters may be calculated from the proportion of the total landings made by vessels, $p(\text{vsl})$, and the proportion of the vessel landings from U.S. waters, $p(\text{US}|\text{vsl})$:

$$(1) \, p(\text{US}) = 1 - p(\text{vsl}) + p(\text{US}|\text{vsl}) \times p(\text{vsl})$$

Values for $p(\text{vsl})$ were linearly interpolated between census years until after 1932, when they were fixed at the 1932 level through 1951 (the statistical bulletins published after 1932 no longer reported landings by vessel class). The results are summarized in Table 4.

1934 to 1951.—Camber (1955) provided annual estimates of the proportion of vessel landings that came from Mexico between 1934 and 1951 (based on records kept by the major fish houses in Florida and Alabama, which he considered accurate). He published his statistics under the headings “Pensacola” and “west Florida,” but stated that the data included vessels based in Mobile, which operated much like the Pensacola fleet. He also stated that the number of trips to areas west of the Mississippi River (e.g., the “Galveston Lumps” off Texas) was small until depth recorders became available in the 1950s and the area was better charted. Hence, it appears the bulk of the landings from domestic waters during this period continued to come from east of the Mississippi River. Therefore, equation (1) was applied to Florida and Alabama using Camber’s statistics for west Florida and Pensacola, respectively.

1952 to 1963.—Carpenter (1965) states that complete statistics of red snapper landings from Mexico were collected by the U.S. Fish and Wildlife service during 1954 to 1963, however he provided only Gulf-wide summaries of the proportion taken from foreign waters. Similar Gulf-wide summaries are available after 1958 in the general section of the annual volumes of Fishery Statistics for the United States. We have been unable to locate any documents with the corresponding state-

specific statistics; however we make the following observations:

- (a) The proportions of the landings in Florida and Texas that came from foreign waters were relatively constant from 1963 to 1966 and are similar to the proportions in given by Camber (1955). The landings for Alabama were not classified by water body in 1963, but the proportion from foreign waters can be calculated from the corresponding proportions for the other states and the Gulf-wide summary. The resulting value, 66%, is also similar to the proportions given by Camber (1955).
- (b) The landings in Mississippi were low and exclusively from U.S. waters until 1957, when they began to increase rapidly. An average of only 26% of the catch was taken in U.S. waters during 1963 (see below)
- (c) The landings in Louisiana were small and there are no historical accounts of any foreign catch during this time (see below)

Based on these observations, linear transitions are assumed for the U.S. proportions (F) for Alabama (from 38% in 1951 to 34% in 1963), Mississippi (from 100% in 1956 to 26% in 1963) and Louisiana (100% for all years). The U.S. proportions for Texas and Florida (F_{Tx} and F_{Fl}) were determined from the Gulf-wide statistics using the landings by state (C), the in-

Table 3. Estimates of the catch of vessels over 5 tons and of smaller boats from Collins and Smith (1891), Radcliffe (1920), Sette (1925), and Jarvis (1935).

year	catch by state and operating units									
	W. Florida		Alabama		Mississippi		Louisiana		Texas	
	Vessels	Boats	Vessels	Boats	Vessels	Boats	Vessels	Boats	Vessels	Boats
1890	3092620	376750	51000	0	0	0	250400	0	0	22000
1891	3904342	268600	62375	0	0	0	240500	0	0	4800
1918	6011297	1218871	798400	0	85752	12480	0	60000	1243002	0
1923	7964907	1506360	970000			103618		175000	1008960	
1927	7609494	1703173	1037722	20928		218706		72000	1234026	3280
1928	6492700	1398503	1131245	169277		97328		48000	1019452	35710
1929	5797225	1902918	1184488	43113		90864		80000	732337	71803
1930	3983215	1018457	820441	27477		188725		76400	835136	94442
1931	3132965	1260027	849451	13750		68236		78835	650827	39837
1932	3096876	1442589	612938	68635		36812		66884	895637	89654

Table 4. Fraction of the total landings made in U.S. waters, $p(US)$, as deduced from the fraction of the total landings made by vessels, $p(vsl)$, and the fraction of vessel landings that came from U.S. waters $p(US|vsl)$. Landings from boats (under 5 tons) are assumed to have come from U.S. waters. Shaded regions represent interpolations or assumptions that are not based on published accounts.

Year	West Florida			Alabama			Texas		
	$p(vsl)$	$p(US vsl)$	$p(US)$	$p(vsl)$	$p(US vsl)$	$p(US)$	$p(vsl)$	$p(US vsl)$	$p(US)$
1889	0.89	1.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00
1890	0.94	1.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00
1891	0.94	1.00	1.00	1.00	1.00	1.00	0.08	1.00	1.00
1892	0.94	1.00	1.00	1.00	1.00	1.00	0.17	1.00	1.00
1893	0.95	1.00	1.00	1.00	1.00	1.00	0.25	1.00	1.00
1894	0.95	1.00	1.00	1.00	1.00	1.00	0.33	1.00	1.00
1895	0.96	0.95	0.95	1.00	0.95	0.95	0.42	0.95	0.98
1896	0.96	0.91	0.91	1.00	0.91	0.91	0.50	0.91	0.96
1897	0.97	0.86	0.86	1.00	0.86	0.86	0.58	0.86	0.92
1898	0.97	0.81	0.82	1.00	0.81	0.81	0.67	0.81	0.87
1899	0.97	0.77	0.78	1.00	0.77	0.77	0.75	0.77	0.83
1900	0.98	0.72	0.73	1.00	0.72	0.72	0.83	0.72	0.77
1901	0.98	0.67	0.68	1.00	0.67	0.67	0.92	0.67	0.70
1902	0.99	0.63	0.63	1.00	0.63	0.63	1.00	0.63	0.63
1903	0.98	0.58	0.59	1.00	0.58	0.58	1.00	0.58	0.58
1904	0.97	0.53	0.54	1.00	0.53	0.53	1.00	0.53	0.53
1905	0.96	0.48	0.50	1.00	0.48	0.48	1.00	0.48	0.48
1906	0.95	0.44	0.47	1.00	0.44	0.44	1.00	0.44	0.44
1907	0.94	0.39	0.43	1.00	0.39	0.39	1.00	0.39	0.39
1908	0.93	0.34	0.39	1.00	0.34	0.34	1.00	0.34	0.34
1909	0.92	0.30	0.36	1.00	0.30	0.30	1.00	0.30	0.30
1910	0.91	0.25	0.32	1.00	0.25	0.25	1.00	0.25	0.25
1911	0.90	0.25	0.33	1.00	0.25	0.25	1.00	0.25	0.25
1912	0.89	0.25	0.33	1.00	0.25	0.25	1.00	0.25	0.25
1913	0.88	0.25	0.34	1.00	0.25	0.25	1.00	0.25	0.25
1914	0.87	0.25	0.35	1.00	0.25	0.25	1.00	0.25	0.25
1915	0.86	0.25	0.36	1.00	0.25	0.25	1.00	0.25	0.25
1916	0.85	0.25	0.36	1.00	0.25	0.25	1.00	0.25	0.25
1917	0.84	0.25	0.37	1.00	0.25	0.25	1.00	0.25	0.25
1918	0.83	0.25	0.38	1.00	0.25	0.25	1.00	0.25	0.25
1919	0.83	0.25	0.38	1.00	0.25	0.25	1.00	0.25	0.25
1920	0.84	0.25	0.37	1.00	0.25	0.25	1.00	0.25	0.25
1921	0.84	0.25	0.37	1.00	0.25	0.25	1.00	0.25	0.25
1922	0.84	0.25	0.37	1.00	0.25	0.25	1.00	0.25	0.25
1923	0.84	0.25	0.37	1.00	0.25	0.25	1.00	0.25	0.25
1924	0.83	0.25	0.38	1.00	0.25	0.25	1.00	0.25	0.25
1925	0.83	0.25	0.38	0.99	0.25	0.26	1.00	0.25	0.25
1926	0.82	0.25	0.39	0.99	0.25	0.26	1.00	0.25	0.25
1927	0.82	0.25	0.39	0.98	0.25	0.26	1.00	0.25	0.25
1928	0.82	0.25	0.38	0.87	0.25	0.35	0.97	0.25	0.28
1929	0.75	0.25	0.44	0.96	0.25	0.28	0.91	0.25	0.32
1930	0.80	0.25	0.40	0.97	0.25	0.27	0.90	0.25	0.33
1931	0.71	0.25	0.47	0.98	0.25	0.26	0.94	0.25	0.29
1932	0.68	0.25	0.49	0.90	0.25	0.33	0.91	0.25	0.32
1933	0.68	0.50	0.66	0.90	0.21	0.29	0.91	0.50	0.55
1934	0.68	0.43	0.61	0.90	0.17	0.25	0.91	0.43	0.48
1935	0.68	0.45	0.62	0.90	0.18	0.26	0.91	0.45	0.50
1936	0.68	0.48	0.65	0.90	0.19	0.27	0.91	0.48	0.53
1937	0.68	0.50	0.66	0.90	0.20	0.28	0.91	0.50	0.55

Table 4. Continued.

1938	0.68	0.58	0.71	0.90	0.20	0.28	0.91	0.58	0.62
1939	0.68	0.65	0.76	0.90	0.19	0.27	0.91	0.65	0.68
1940	0.68	0.58	0.71	0.90	0.18	0.26	0.91	0.58	0.62
1941	0.68	0.49	0.65	0.90	0.18	0.26	0.91	0.49	0.54
1942	0.68	0.52	0.67	0.90	0.21	0.29	0.91	0.52	0.56
1943	0.68	0.44	0.62	0.90	0.18	0.26	0.91	0.44	0.49
1944	0.68	0.47	0.64	0.90	0.17	0.25	0.91	0.47	0.52
1945	0.68	0.42	0.60	0.90	0.15	0.23	0.91	0.42	0.47
1946	0.68	0.53	0.68	0.90	0.26	0.34	0.91	0.53	0.57
1947	0.68	0.55	0.69	0.90	0.24	0.32	0.91	0.55	0.59
1948	0.68	0.52	0.67	0.90	0.26	0.34	0.91	0.52	0.56
1949	0.68	0.57	0.71	0.90	0.25	0.32	0.91	0.57	0.61
1950	0.68	0.52	0.67	0.90	0.20	0.28	0.91	0.52	0.56
1951	0.68	0.63	0.75	0.90	0.31	0.38	0.91	0.63	0.66

terpolated fractions for Louisiana, Mississippi and Alabama described above, and assuming the ratio of the fractions for Texas and Florida r was constant at the 1963–1966 average (0.89):

$$(2) \quad F_{Fl,y} = \frac{F_{total,y}C_{total,y} - F_{Al,y}C_{Al,y} - F_{Ms,y}C_{Ms,y} - C_{La,y}}{C_{Fl,y} + rC_{Tx,y}}$$

$$F_{Tx,y} = rF_{Fl,y}$$

The results of these calculations are summarized in Table 5.

As mentioned previously, the statistics published between 1932 and 1962 do not include estimates of the fraction of the landings in Florida and Alabama that came from east or west of the Mississippi River. Camber (1955) states that in recent years (presumably the early 1950s) appreciable quantities of red snapper were caught on the “western grounds,” located near the 100 fathom lime (200 m) off Texas and Louisiana. He also mentions that vessels from Pensacola and Mobile fished in this area (as well as vessels from Galveston). In the absence of any other information, it is assumed that the fraction of the catch taken by eastern-based fleets from western waters since 1951 increased linearly to the mean fraction from the years 1963–1966 (30% for Florida and 9% for Alabama, see Table 6).

Mississippi

Radcliffe (1920) and Jarvis (1935) indicate that the snapper boats operating out of Mississippi were relatively small and fished exclusive-

ly in U.S. waters. This appears to have remained the case through the early 1950s; Camber (1955, p. 14) indicates that the main Mississippi fleet (out of Pascagoula) did not fish on the major foreign fishing grounds. However, during the late 1950s the landings of red snapper in Mississippi began to increase rapidly, apparently due to an increase in the number of vessels fishing in foreign waters (the U.S. fishery statistics for 1963–1966 indicated that only 27% of the landings in Mississippi were taken from U.S. waters). For lack of more detailed information, we assume that the proportion of Mississippi landings from U.S. waters decreased linearly from 100% in 1956 to 26% by 1963.

Vessels operating out of Pascagoula are cited by Camber (1955) as fishing on both sides of the river, but no percentages are given. The 1963–1966 U.S. fishery statistics indicate that 95% of the Mississippi landings from U.S. waters were taken west of the river. It is unclear that an equally large fraction applies to the earlier years of the fishery, but the Mississippi landings are small until the mid 1950s and any errors introduced by applying this fraction should have a negligible impact on the final catch series.

Louisiana

The landings of red snapper in Louisiana, like those in Mississippi, were relatively low and of local origins until the late 1950s, after which time they increased rapidly. However, unlike Mississippi, the increase in landings does not appear to

Table 5. Fraction of the landings in each state that came from U.S. waters and for all states combined (Total). Shaded regions represent values based on interpolations or assumptions that are not based on reliable published accounts. The values in bold (TX and FL) were inferred using equation (2).

year	TX	LA	MS	AL	FL	Total
1951	0.66	1.00	1.00	0.38	0.75	0.67
1952	0.56	1.00	1.00	0.38	0.63	0.58
1953	0.46	1.00	1.00	0.38	0.52	0.49
1954	0.36	1.00	1.00	0.37	0.41	0.40
1955	0.34	1.00	1.00	0.37	0.38	0.39
1956	0.38	1.00	1.00	0.37	0.43	0.44
1957	0.42	1.00	0.89	0.36	0.48	0.48
1958	0.67	1.00	0.79	0.36	0.76	0.70
1959	0.64	1.00	0.68	0.36	0.73	0.65
1960	0.73	1.00	0.58	0.35	0.82	0.70
1961	0.66	1.00	0.47	0.35	0.75	0.64
1962	0.70	1.00	0.37	0.35	0.79	0.65
1963	0.74	1.00	0.26	0.34	0.66	0.57

have been the result of increased fishing in foreign waters. The fleet appears to have expanded offshore into so-called “western grounds” near the 100 fathom line. The U.S. fishery statistics for 1963–1966 indicate that all of the landings in Louisiana were caught in U.S. waters. Hence, it appears safe to assign all Louisiana landings to U.S. waters west of the Mississippi River.

Texas

No statistics are available on the proportion of the landings in Texas that came from foreign waters prior to 1963. Smith (1894, 1895) reported that three vessels from Galveston went fishing on the Campeche Banks in 1892, but the venture was plagued by an inadequate supply of ice and difficulties in securing labor. Camber (1955) indicated that, at the turn of the century, most of the landings in Texas were still being made by smaller vessels operating in local waters. Camber (1955, p. 39) does provide statistics for the number of vessels fishing on the Campeche Banks in 1951, which when combined with data on the total number of vessels in the 1951 U.S. fishery statistics bulletin, suggests that the proportion of vessels that fished regularly off the Campeche Banks was about the same for Texas (5 or 6 of 18) as for Florida (24 of 75). The proportions of the landings from foreign waters during 1963–1966 are also quite similar between

the two states (about 38% for Texas and 30% for Florida). Based on these observations, we apply the proportions obtained for Florida (above) to Texas through 1951. For 1952 to 1962 we make the somewhat less restrictive assumption that the ratio of the proportions for Texas and Florida are constant (at the 1963–1966 level) and apply the method discussed in connection with equation (2) above. We recognize that neither assumption is especially well supported, however the landings in Texas during this period were typically on the order of 15% of the total, so we do not expect the error in this regard to be large.

Camber (1955) indicates that waters east of the Mississippi River were fished exclusively by boats operating out of ports from that region (e.g., Pascagoula, Mobile, Pensacola, Panama City, Tampa) and that the waters west of the Mississippi River were fished by vessels operating out of Galveston, Freeport, Brownsville and Pascagoula. This is consistent with the 1963–1966 U.S. fishery statistics, which indicate that 100% of the landings in Texas and Louisiana from U.S. waters were taken west of the Mississippi.

Other Adjustments

Two other adjustments were made to the landings data, one to account for cases where the landings were recorded in dressed (gutted)

weight rather than whole weight and another to account for the fraction of the red snapper landings that were actually *L. campechanus*.

Conversion from dressed weights

Camber (1955) reported that red snapper were landed whole prior to 1934 and dressed after that. However, the descriptions in the various U.S. fishery statistics bulletins (e.g., Anderson and Peterson 1953; p. 461) generally indicate that the dressed landings were converted to whole weight assuming a whole weight to dressed weight ratio of 1.11. The only exception appears to be Florida, in which case the landings from 1956 to 1985 were never converted from dressed to whole weight (E. Snell, SEFSC, personal communication).

Correction for species composition

Dealers frequently marketed other species as “red snapper,” including silk snapper (yellow eye snapper, *L. vivanus*), gray snapper *L. Griseus*, lane snapper *L. synagris*, mutton snapper *L. analis*, and vermilion snapper *Rhomboplites aurorubens*. However, Camber (1955) noted that typically 95% or more of the landings were in fact *L. campechanus* (then called *L. aya*). He adjusted the landings statistics of Florida producers from 1880 to 1951 based on his review of fish house records and samples of the catch. His results are summarized in Table 7. Here it assumed with no further information that these fractions apply equally to the landings in other areas. The value of 96% is assumed to continue through 1962 inasmuch as port agents apparently did not make a concerted effort to distinguish the red snapper catch

by species until 1986 (G. Davenport, SEFSC, personal communication.).

Results and Discussion

Estimates of the annual landings of red snapper caught in the U.S. Gulf of Mexico (east and west of the Mississippi River) were obtained by multiplying four quantities: the total landings in whole weight (Table 2), the proportion from U.S. waters (Tables 4 and 5), the proportion of landings from U.S. waters that came from east or west of the Mississippi River (Table 6), and the fraction of the landings sold as red snapper that actually were *L. campechanus*. The results are shown in Table 7 and Figure 2.

The landings from waters west of the Mississippi River appear to have been relatively small, generally under 1 million pounds, until the late 1940s. The landings from waters east of the Mississippi River, on the other hand, show a strong increasing trend that reflects the initial expansion of the fishery from a few vessels fishing locally off Pensacola in 1872 to dozens of vessels fishing as far as the Dry Tortugas by 1902. After 1902 the landings from the east appear to have diminished rapidly, primarily in response to a shift in effort to the Campeche Banks (Figure 1). The landings from the east increased somewhat during the early 1920's owing to an increase in the number of smaller boats (<5 tons) fishing locally and the widespread use of steam engines. The landings dropped again during the 1930s, which is consistent with anecdotal accounts of a decline in market demand during the Great Depression (Camber 1955). The landings began to increase during the mid 1930s as the economy recovered, but dipped once again during World War II when many fishermen enlisted

Table 6. Fraction of landings in each state that came from the adjacent region (i.e., from the same side of the Mississippi River as the port was located). Here the state of Mississippi is considered to lie in the eastern region.

Year	TX	LA	MS	AL	wFL
1872-1891	1.00	1.00	0.05	1.00	1.00
1892-1949	1.00	1.00	0.05	1.00	1.00
1950-1962	1.00	1.00	0.05	0.91	0.70

Table 7. A reconstructed history of red snapper landings (in lbs) from U.S. waters.

year	Red Snapper and others		Proportion <i>L. campechanus</i> (Gulf-wide)	Red Snapper (<i>L. campechanus</i>)	
	East	West		East	West
1872	423796	0	0.99	419574	0
1873	671009	0	0.99	664325	0
1874	918223	0	0.99	909076	0
1875	1165438	0	0.99	1153828	0
1876	1377335	0	0.99	1363615	0
1877	1165438	0	0.99	1153828	0
1878	1059489	0	0.99	1048934	0
1879	1165438	0	0.99	1153828	0
1880	1843284	900000	0.99	1824922	891034
1881	3033916	810000	0.99	3003739	801943
1882	3643373	719000	0.99	3607188	711859
1883	3348190	640667	0.99	3314988	634313
1884	3764275	562333	0.99	3727003	556765
1885	3335290	483000	0.99	3302316	478225
1886	4001354	404667	0.99	3961855	400672
1887	4042362	206000	0.99	4002518	203970
1888	3310000	215000	0.99	3277425	212884
1889	3520369	272000	0.99	3485777	269327
1890	4234942	245000	0.99	4192270	242531
1891	4348585	276000	0.99	4304834	273223
1892	4577308	307000	0.99	4531325	303916
1893	4806030	339000	0.99	4757824	335600
1894	5035753	370000	0.99	4985319	366294
1895	4896949	394839	0.99	4847980	390891
1896	4571590	414865	0.99	4525762	410706
1897	4880880	427242	0.99	4831831	422949
1898	5423822	685942	0.99	5369317	679049
1899	6108691	915215	0.99	6047304	906018
1900	6589836	1095365	0.99	6523614	1084358
1901	6963091	1216611	0.99	6893118	1204385
1902	7308770	1302840	0.99	7235733	1289821
1903	6640382	1217420	0.99	6574083	1205265
1904	6008148	1128370	0.99	5948213	1117114
1905	5404274	1036800	0.99	5350410	1026466
1906	4930276	964040	0.99	4881180	954440
1907	4378625	866190	0.99	4335060	857572
1908	3853806	765680	0.99	3815496	758069
1909	3447084	660812	0.99	3412777	654236
1910	2972403	543525	0.99	2942785	538109
1911	2969664	532836	0.99	2940039	527520
1912	2966329	523098	0.99	2936702	517874
1913	2962939	513611	0.99	2933312	508475
1914	2958322	503873	0.99	2928706	498829
1915	2953059	494136	0.99	2923461	489183
1916	2947103	483447	0.99	2917530	478596
1917	2940801	473709	0.99	2911256	468950
1918	2925910	463972	0.99	2896481	459305
1919	3111272	476173	0.99	3079966	471382

Table 7. (Continued)

1920	3228246	488374	0.99	3195750	483458
1921	3402973	501777	0.99	3368705	496724
1922	3577532	513978	0.99	3541492	508800
1923	3745059	526179	0.99	3707316	520876
1924	3727033	508289	0.99	3689539	503176
1925	3639188	490399	0.99	3602644	485474
1926	3608224	472259	0.99	3572054	467525
1927	3896569	591829	0.99	3857579	585907
1928	3478754	431155	0.99	3444187	426871
1929	3695890	421321	0.99	3658800	417093
1930	2256050	559149	0.99	2233495	553559
1931	2272708	346288	0.99	2249781	342794
1932	2440472	415464	0.99	2416037	411305
1933	3025882	591087	0.99	2995666	585185
1934	2637816	502370	0.99	2611546	497367
1935	2994761	696661	0.97	2902548	675210
1936	3394723	904123	0.95	3219475	857449
1937	3341912	1059607	0.94	3139239	995347
1938	4096645	1041120	0.97	3973613	1009853
1939	4460916	917750	0.97	4326942	890188
1940	3107244	891646	0.97	3001832	861397
1941	3038129	825124	0.96	2917622	792396
1942	2386241	594977	0.98	2329540	580839
1943	2006007	415111	0.97	1953460	404237
1944	2263894	309890	0.97	2206368	302015
1945	2039656	173169	0.97	1985188	168545
1946	3204031	357116	0.96	3086345	343998
1947	3276483	524804	0.96	3161005	506307
1948	3655589	658640	0.96	3520166	634241
1949	4103870	941437	0.97	3965333	909656
1950	2316791	1792748	0.96	2230589	1726045
1951	2695334	1750439	0.96	2586570	1679804
1952	2955945	2001141	0.96	2836665	1920390
1953	2369639	1416738	0.96	2274018	1359569
1954	2070236	1315562	0.96	1986697	1262476
1955	2262983	1461874	0.96	2171666	1402884
1956	2574421	1854330	0.96	2470537	1779504
1957	2648857	2112996	0.96	2541969	2027731
1958	4357859	3532154	0.96	4182008	3389623
1959	4029521	3535484	0.96	3866920	3392818
1960	4471641	3765292	0.96	4291199	3613353
1961	4145286	4391495	0.96	3978014	4214287
1962	3945536	4124420	0.96	3786324	3957989
1963	3906025	3709840	0.96	3748408	3560139

U.S. landings from U.S. Gulf of Mexico

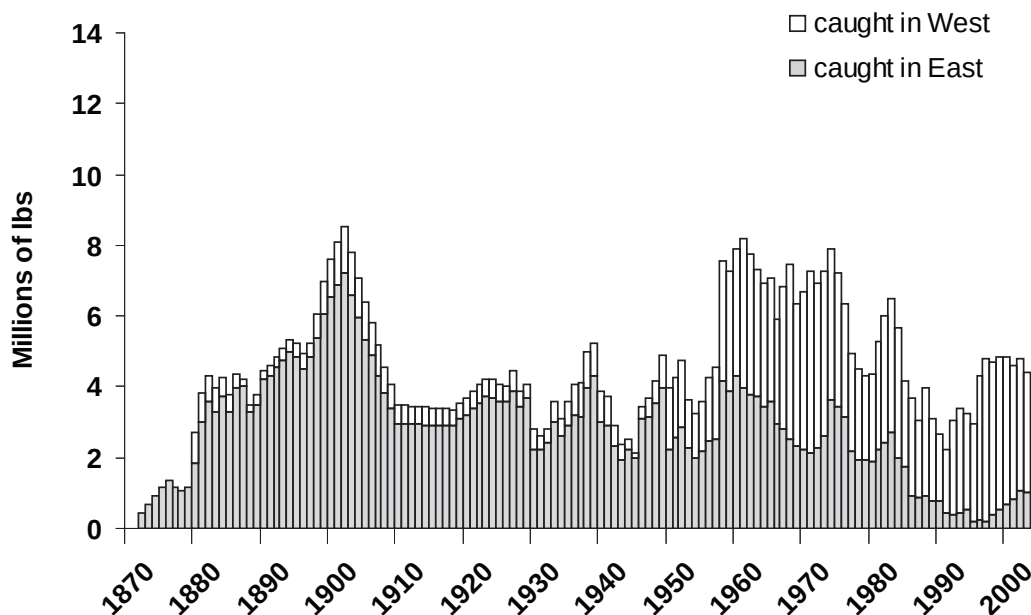


Figure 2. Reconstructed landings of red snapper caught in the U.S. Gulf of Mexico (east and west of the Mississippi River).

and the remaining crews were forced to fish in shallow waters to avoid the threats of marauding U-boats (Camber 1955). The landings from both sides of the River, but especially the west, increased rapidly after the war owing in part to several technological innovations such as the fathometer, reels, and wire line, which opened up new fishing grounds in deeper water (Siebenaler and Brady 1952). The landings continued to increase through the late 1950s and early 1960s, presumably owing to what Carpenter (1965) described as a tremendous increase in the size of the commercial fleet.

The expansiveness of the shaded regions in Tables 4 through 7 attests to the substantial gaps in the historical record of Gulf of Mexico red snapper landings. As discussed above, a number of interpolations or other assumptions had to be made which were not directly grounded in the U.S. census data or other reliable published accounts. In many cases even severe violations of these assumptions would have little impact on the overall result because the landings from the affected strata are negligible (e.g., Louisiana and Mississippi prior to the late 1950s). In other cases, the fisheries involved may not have

changed their mode of operation for an extended time period, so the interpolated values are relatively constant and errors are likely to be relatively small. Nevertheless, there are several gaps which are of particular concern.

Perhaps the greatest uncertainties lie in the magnitude of the landings during the many uncanvassed years between 1890 and the 1920s, and in the fraction of those landings that came from the Campeche Banks. Here the total landings were linearly interpolated between years with census estimates, which seems consistent with anecdotal accounts of a rapidly growing fishery during the early years. After 1902, however, the landings appear to have declined somewhat. Jarvis (1935) states that severe hurricanes in 1906, 1916, 1917, and 1926 damaged the fleet and resulted in reduced catches; one firm alone apparently losing 14 of its vessels in the 1906 storm. In light of this, it is probably fair to say that the interpolation approach correctly reflects the overall decline between 1902 and the 1920s, but overestimates the landings during the years affected by hurricanes (to a degree we cannot reasonably surmise). Of equally uncertain status is the nature of the transition from negligible for-

eign catches prior to 1900 to about 75% foreign in 1910 (and constant at that level thereafter). The anecdotal accounts we have found seem to confirm that the move to Campeche was neither abrupt nor complete owing to the expense of ice, inexperience of the captains in uncharted waters and disputes over the rights of American vessels to fish in waters claimed by Mexico (e.g., Bortone et al. 1997). Nevertheless, there is no particular reason to assume the transition was precisely linear and the question remains of some importance given that the landings at that time are among the highest on record.

Also of particular concern are the landings in Texas after about 1900. The references we have found so far do not give much indication of the fraction of the landings that came from foreign waters. It is known that at least three vessels from Galveston, Texas were fishing off the Campeche Banks as early as 1892 (Smith 1894) and that by 1897 vessels from Galveston were fishing off Mexico on a regular basis (Carpenter 1965). This, plus the observation that the trends in Florida and Texas were similar during the 1950s and 1960s, is essentially all the evidence we have to back our assumption that percentage of the Texas landings coming from foreign waters was roughly the same as for Florida. It is of interest to note, however, that the total landings in Texas were generally under 2 million pounds for most of its recorded history; far less than recent levels. Hence, it seems unlikely that errors in this regard would have an especially great effect on ones perception of the fishery or the results of the SEDAR stock assessment (Porch 2007, this volume).

We recommend that subsequent research focus on locating additional observations relating to the fraction of the landings coming from foreign waters. In this regard we note that the landings statistics for a given year are often summarized to various degrees in multiple documents. For example, the water body where the catches were made appears in the NMFS electronic database, but not in 1963 Fishery Statistics summaries (Lyles 1965). In addition, we sometimes have the fraction of the landings that came from foreign waters on a Gulf-wide basis, in which case state-specific estimates must also have been made. This suggests to us that more

complete statistics were generally collected, but simply not published in the various digests we have available to us. Perhaps the more detailed data still exists somewhere in State and Federal archives.

Finally, as mentioned earlier, past stock assessments of Gulf red snapper have generally focused on the years after 1962, when detailed landings are available. However, it was found that the lack of contrast in those data made it difficult to reliably estimate measures of stock productivity such as MSY. Comparisons made during SEDAR 7 showed that the estimates of stock abundance obtained with the extended series (1872–2003) were rather similar to those obtained using only the more recent data (1963–2003). However, the estimates of MSY (conditioned on recent fishing patterns) obtained with the shorter time series were highly uncertain and more than three times larger than the corresponding estimates from the extended time series (over 20 mp compared with about 6 mp). The high MSY value obtained for the shorter time series was viewed as unlikely because the stock was estimated to be overfished even though the historical landings have probably never approached 20 mp. For these and other reasons, the participants in SEDAR 7 elected to base their advice on the models that employed the extended time series developed here (for more details see Cordue 2005; SEDAR 2005; Porch 2007, this volume).

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An Assessment of the Red Snapper Fishery in the U.S. Gulf of Mexico Using a Spatially-Explicit Age-Structured Model

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Abstract.—Red snapper *Lutjanus campechanus* have been fished in the Gulf of Mexico since before the Civil War. The size and efficiency of the commercial fleet increased greatly during the 1960s, but without a corresponding increase in catch, suggesting that red snapper populations throughout the Gulf of Mexico were by that time fully-exploited and perhaps even overfished. Nevertheless, most assessments of red snapper in the U.S. Gulf of Mexico have been based on data collected since the 1980s owing to a combination of gaps in the catch data and limitations of the models employed. The lack of contrast in the more recent data makes it difficult to develop meaningful estimates of stock status, particularly in relation to abundance-based reference points such as the equilibrium spawning biomass at maximum sustainable yield. This paper presents a flexible age-structured model that includes information dating back to the inception of the fishery. The results suggest that the populations of red snapper in the U.S. Gulf of Mexico are well below the levels corresponding to a spawning potential ratio of 30%. They also suggest the stock will not recover to that level in the foreseeable future without substantial reductions in both the catch of adults by the directed fleets and the bycatch of juveniles by the offshore shrimp fishery.

Introduction

Red snapper *Lutjanus campechanus* have long been esteemed as one of the finest food-fishes in the Gulf of Mexico. The first fisheries for red snapper developed during the late 1840s to supply local markets in New Orleans (Louisiana) and Mobile (Alabama), but the trade was interrupted by the Civil War (Bortone et al. 1997). The fishery was reestablished after the war by several persistent New Englanders who found that packing red

snapper on ice would keep them fresh long enough to ship to lucrative northern markets. Production on a commercial scale appears to have begun in 1872 with the formation of the Pensacola Fish Company (Warren 1898). From there the fishery expanded rapidly, fueled by the growing demand as more and more markets were reached through the improving railroad system. By the turn of the century record landings were being made by dozens of vessels operating throughout the Gulf of Mexico. As one dealer put it, “No man who

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is willing to buy a red snapper has lacked the opportunity” (Jordan and Evermann 1923).

Troubling signs that all was not well with the red snapper fishery were evident as early as the 1880s. Stearns (1883), for example, remarked that “most of the old fishing grounds are barren, and smacks have to go farther each year to find new ones.” Collins (1885) went so far as to say “...it is probable that this species is being more or less rapidly depleted, and the day may not be far distant when the services of the Commission will be needed to keep up the supply, if not to prevent the practical destruction of the important fishery now prosecuted for red snapper.” Concerns about depletion and overfishing have echoed through the Gulf ever since.

Several technological innovations introduced after World War II (diesel engines, depth recorders, wire line, and reels) allowed vessels to fish farther away and in deeper waters than ever before. Nevertheless, by the early 1950s most of the snapper banks off west Florida were considered impoverished and catch per unit effort was observed to be declining on the Campeche Banks (Camber 1955). Continued declines in catch per unit effort and in the size of the fish caught raised concerns among fishermen, who attributed the declines to an increase in the number of snapper vessels and an increase in the bycatch of juvenile snapper by shrimp trawlers (Moe 1963; Lyles 1965, Bradley and Bryan 1975). A further blow was dealt during the 1980s when the U.S. fleet was excluded from the Campeche Banks by the Mexican government and had to redirect its effort towards domestic waters.

The first quantitative analysis of the red snapper population in the Gulf of Mexico was conducted by Nelson and Manooch (1982). They estimated fishing mortality rates on the order of 0.6 year^{-1} off Louisiana and 0.23 year^{-1} off West Florida (based on catch-curves applied to age samples from the commercial landings between 1978 and 1979). The estimates for Louisiana in particular were several times greater than commonly assumed levels of adult natural mortality (0.1 to 0.2 year^{-1}) and could have been interpreted as evidence of overfishing. Goodyear (1988), using data collected between 1984 and 1986, showed that fishing mortality was not constant with age and probably declined from about 0.75

at age 2 to about 0.3 at age 8 or 9. He also estimated spawning biomass to be less than 5% of unfished levels. At about the same time, Nichols et al. (1987) estimated that upwards of 12 million juvenile (age 0 and age 1) red snapper were killed as a bycatch of offshore shrimp trawlers and Powers et al. (1987) showed that eliminating this bycatch could increase the long term yield to the red snapper fishery by as much as 90%. As a result of these and subsequent studies, the fishery was declared overfished and a series of regulatory amendments were introduced to help rebuild the stock (see Hood et al. 2007, this volume).

Several stock assessments were conducted by the National Marine Fisheries Service during the 1990s, the latest of these being Schirripa and Legault (1999). Each indicated that the stock remained in an overfished condition and that both the directed harvest and trawl bycatch needed to be reduced substantially. Nevertheless, the results were generally regarded as highly uncertain and managers often opted for total allowable catch limits (TACs) that were at or above the maximum recommended levels (Hood et al. 2007, this volume). Much of the uncertainty can be attributed to the fact that the assessments were based only on recent data collected since the 1980s, when red snapper populations are likely to have already been depressed. The lack of contrast in such data, coupled with the lack of historical perspective, makes it difficult to develop meaningful estimates of stock status, particularly in relation to abundance-based reference points such as maximum sustainable yield.

The restriction to data collected since the 1980s is a consequence of both gaps in the available historical record and limitations of the methodology employed. The former has partly been addressed by Porch et al. (this volume), who have supplemented the extant landings database with observations gleaned from historical documents. The latter can be addressed by the use of modern age-structured statistical algorithms that can accommodate missing data and incorporate auxiliary data sources that cannot be tapped by simpler methodologies. The purpose of this paper is to present the age-structured statistical algorithm CATCHEM_AD; its math-

Table 1. Method of accounting used in the stock assessment algorithm. The entries represent a cohort, with cohort 1 being born in season 1 of year 1, cohort 2 being born in season 2 of year 1, and so on. In this example there are three years of data with each year having two seasons, and therefore there are six seasonal age-classes. Thus, in order to have a complete age-structure by the first season of the data period (season 1 of year 4), it is necessary to track the first five cohorts recruited immediately prior to the data era.

		Prehistoric era						Data era						Future era		
Year		1	1	2	2	3	3	4	4	5	5	6	6	7	8...	
Season		1	2	1	2	1	2	1	2	1	2	1	2	1	2...	
Age	1	1	2	3	4	5	6	7	8	9	10	11	12	13	14...	
	2			1	2	3	4	5	6	7	8	9	10	11	12	13...
	3				1	2	3	4	5	6	7	8	9	10	11	12...
	4					1	2	3	4	5	6	7	8	9	10	11...
	5						1	2	3	4	5	6	7	8	9	10...
	6							1	2	3	4	5	6	7	8	9...

ematical underpinnings and two particular applications of it that provided the basis for much of the management advice generated during the latest assessment (SEDAR 2004; Cordue 2005). As such, the reader should appreciate that the paper covers only a portion of the work that was presented during the lengthy SEDAR assessment process and is not intended to supersede that assessment. In addition, certain features that were not used in the original assessment, but which may become useful in the future, are discussed in anticipation of forthcoming information on red snapper movement and recruitment patterns.

Model Structure

The basic units in the model are stock, cohort, fleet, habitat, era, year, season and age-class. As used here, a “stock” designates any population of animals with similar life history characteristics for which a distinct accounting is desired. For example, a stock may be identified with a growth-morph, sex, or species. The term “cohort” is used to refer to members of a given stock that were born during the same year and season. A “fleet” refers to any entity that harvests or samples any of the stocks (e.g., a fishery, scientific survey or predator species). The term “habitat” represents any form of

spatial domain where the concentration of the stock or fleet may vary in important ways from the overall mean density. The time unit “era” is used to distinguish three periods of exploitation: a “prehistoric” era, during which no data are available; a “data” era, when presumably there are data useful for estimating abundance and mortality; and a “future” era, when mortality rates are assumed (input). The duration of the prehistoric era is set equal to the number of seasonal age-classes so as to generate a complete age-structure by the beginning of the first year of the data era. The time units “year” and “season” are used in the conventional sense, except that the number of seasons per year may be defined by the user. The calculations are done on a seasonal basis to accommodate seasonal movement and fishing patterns and to mimic the effect of temporally protracted spawning by allowing for multiple cohorts per year. The model tracks the abundance of each cohort throughout its life span as shown in Table 1.

The age-classes range from 1 to A , where r is the age (in seasons) associated with age-class 1 and subsequent age-classes are incremented forward by one season. The last age-class, A , is not cumulative, i.e., fish are assumed to have a maximum life span of $A + r - 1$ seasons. The calendar year y and season s are inferred from

the cohort c and age-class a as

$$(1) \quad y\{c, a\} = y_0 + \text{int}\left\{\frac{c+a-2}{n\{s\}}\right\}$$

$$s\{c, a\} = c + a - 1 - \text{int}\left\{\frac{c+a-2}{n\{s\}}\right\}n\{s\}$$

where y_0 is the first year of the prehistoric period, $\text{int}\{x\}$ is the integer portion of quantity x and $n\{s\}$ is the number of seasons in a year. Hereafter, the notation $\{c, a\}$ will be omitted for compactness, with the implicit understanding that s and y are derived quantities. Otherwise, curly braces are used throughout to distinguish function arguments from calculation precedence.

Population Dynamics Model

The progression from one age-class to the next is modeled as

$$(2) \quad N_{cah} = \sum_j \tilde{N}_{caj} T(h|j, a, s)$$

$$(3) \quad \tilde{N}_{c, a+1, h} = \begin{cases} R_c \tau_h & a = 0 \\ N_{cah} e^{-Z_{asyh}} & 1 \leq a < A \\ 0 & a \geq A \end{cases}$$

where the subscript h (or j) identifies habitat. The subscript indexing stock has been omitted for convenience of notation, but the equations should be understood to depend on stock as well. The variable R_c denotes the initial recruitment to cohort c , τ_h denotes the probability that a new recruit will start out in habitat h , and Z denotes the instantaneous total mortality rate. The variables $\tilde{N}_{cah}$ and N_{cah} denote the number of survivors in habitat h before and after a movement event, which is assumed to occur instantaneously at the beginning of the season (T denotes the transfer probability described below). Note that the movements of red snapper between habitats were not modeled in this paper owing to insufficient data. However, a brief description of the underlying model is provided here in anticipation of future applications with more informative data.

Movement.—The conditional probability T that a fish will transfer to habitat h given its cur-

rent location j is modeled as a diffusive process where the net pull towards a given habitat is a function of the difference between the intrinsic attraction of a habitat (β_1) and the difficulty in getting to it (β_2):

$$(4) \quad T(h|j) \propto e^{-(\beta_2 - \beta_1)}$$

Specifically, β_1 is expressed as a categorical variable that varies by habitat, age-class and season and β_2 is expressed as the effective distance between habitats (x_{hj}) divided by the diffusion velocity of each age-class in distance units per season (u_a , which may or may not be proportional to swimming speed):

$$(5) \quad T(h|j, a, s) = \frac{\tau_{ash} e^{-x_{hj}/u_a}}{\sum_i \tau_{asi} e^{-x_{ji}/u_a}}$$

The summation occurs over all possible habitats, ensuring that the all fish are accounted for, i.e., $\sum_h T(h|j, a, s) = 1$ for all j, a, s . Essentially, this is a discrete version of the Joseph and Sendner (1958) diffusion equation immersed in an inhomogeneous advection field. The τ parameters can be thought of as the relative distribution of the cohort among habitats that would be achieved with an infinite diffusion velocity u . Purely diffusive motion is achieved when the τ parameters are identical and the matrix of distance parameters x_{hj} is symmetric ($x_{hj} = x_{jh}$).

Mortality.—The instantaneous mortality rate Z is modeled as the sum of coefficients reflecting natural (M) and fishing-related (F) causes:

$$(6) \quad Z_{asyh} = M_a + \sum_i F_{iasyh}$$

where i indexes a particular source of fishing mortality, hereafter referred to as a fleet. The fishing mortality rate parameters are further decomposed into separable age-dependent and time-dependent effects:

$$(7) \quad F_{iasyh} = q_{iy} v_{ia} f_{iy} \xi_{ia} \delta_{ish} / n\{s\}$$

where q represents the “catchability” of the most vulnerable age-class, v represents the relative vulnerability of the remaining age-classes, f is the total effort exerted by the fleet over all seasons, and ξ is the probability that a fish will

die once it is caught (i.e., the probability it was either landed or suffered post-release mortality). The quantity δ_{ish} equals 1 or 0 depending on whether the fleet does or does not operate in season s or habitat h . Essentially this model assumes that the fishing effort of any given fleet may vary from year to year, but is otherwise spread evenly over the seasons and habitats the fleet is operating. The vulnerability parameters implicitly include the effects of factors such as gear selectivity and the fraction of the stock exposed to the fishery.

Inter-annual variations in f and q are modeled as first-order, lognormal auto-regressive processes, e.g.,

$$(8) \quad f_{iy} = \mu\{f_{iy}\}e^{\varepsilon_{iy}} \\ \varepsilon_{iy} = \rho\{f_{iy}\}\varepsilon_{i,y-1} + \eta_{iy}$$

where μ and ρ represent the median and correlation coefficient of the f_{iy} , respectively, and the η_{iy} are normal distributed random variables with mean zero and standard deviation $\sigma\{f_{iy}\}$. In the present application, the $\mu\{f_{iy}\}$ are model inputs, hopefully based on some index of the relative effort expended by each fishery. For σ sufficiently large, the f_{iy} essentially become free parameters and the values of $\mu\{f_{iy}\}$ become arbitrary. However, the absence of data during the ‘prehistoric period’ generally precludes the estimation of unconstrained changes in effort. Accordingly, for the prehistoric period σ is set to 0, such that $f_{iy} = \mu\{f_{iy}\}$.

Recruitment and the definition of spawning potential.—The recruitment to the first age-class of each cohort (R) is modeled as a first-order, lognormal auto-regressive process,

$$(9) \quad R_c = \mu\{R_c\}e^{\varepsilon_y} \\ \varepsilon_y = \rho\{R_c\}\varepsilon_{y-1} + \eta$$

where μ is the median recruitment, ρ is the correlation coefficient and η is a normal-distributed random variate having mean 0 and standard deviation $\sigma\{R_c\}$ (ostensibly representing the effect on recruitment of fluctuations in the environment). The recruitment standard deviation and correlation coefficient can be difficult to estimate without a good index of recruitment and

may have to be fixed to some moderate values (say $\sigma_R = 0.4$ and $\rho = 0.5$).

The median recruitment can be a constant or specified as truncated Ricker or Beverton–Holt functions that have been recast in terms of the maximum lifetime reproductive rate (α), virgin recruitment during peak season (R_0) and spawning potential relative to virgin levels during peak season (ϕ):

$$(10) \quad \mu\{R_c\} = \begin{cases} R_0\phi_c\alpha^{1-\phi_c} & \text{Ricker} \\ R_0\frac{\alpha\phi_c}{1+(\alpha-1)\phi_c} & \text{Beverton and Holt} \end{cases}$$

(see appendix 1). Note that α is related to the so called “steepness” parameter (ψ , the recruitment when spawning potential is 20% of the unfished level relative to R_0):

$$(11) \quad \psi = \begin{cases} 0.2\alpha^{4/5} & \text{Ricker } (0.2 \leq \psi) \\ \frac{\alpha}{\alpha+4} & \text{Beverton and Holt } (0.2 \leq \psi < 1) \end{cases}$$

The parameters R_0 and α are generally estimated during the model fitting (see below), whereas the spawning potential ϕ is derived from other modeled quantities. In the case of a single unit stock and a single habitat, the calculation of ϕ is straightforward,

$$(12) \quad \phi_c = S_c / S_0 \\ S_c = \sum_a E_{as} N_{\hat{c}a} \quad (\hat{c} = c - r - a + 1) \\ S_0 = R_0 \sum_a E_{as} \exp\left(-\sum_{j=1}^{a-1} M_j\right)$$

Here S_c represents the spawning potential that contributed to cohort c and S_0 is the spawning potential of the unexploited state during the peak spawning season (indexed by the subscript $\hat{s}$). The variable E_{as} represents a measure of the per-capita number of eggs produced by each age-class. The subscript $\hat{c}$ indexes the cohort that was age a at the time of spawning (r seasons prior to the recruitment of cohort c).

When there are multiple stocks and multiple habitats a number of alternatives present themselves. One extreme is to assume that all members of stock k contribute to the net spawning potential of that stock regardless of their current

location, as might occur if the adults generally migrate back to the spawning habitat or the larvae are spatially well-mixed. In that case

$$(13) \quad \begin{aligned} \phi_{ck} &= S_{ck} / S_{0k} \\ S_{ck} &= \sum_a E_{ask} \sum_h N_{\hat{c}ahk} \\ S_{0k} &= R_{0k} \sum_a E_{ask} \exp \left(- \sum_{j=1}^{a-1} M_{jk} \right) \end{aligned}$$

Alternatively, one might wish to model a situation where the members of the various stocks spawn more opportunistically. For example, if all adult fish located in a spawning habitat during the spawning season contribute to the spawning potential of the habitat, regardless of their original stock affiliation, then one obtains:

$$(14) \quad S_{ck} = \sum_a \sum_j E_{asj} N_{\hat{c}ah\{k\}j}$$

where the subscripts j and k are used to index stocks and $\hat{h}\{k\}$ identifies the spawning habitat associated with stock k . The expression for S_{0k} is tedious to write in this scenario, but can be obtained from equations (2) and (3) with recruitment fixed to the stock specific values of R_{0k} and zero fishing mortality. Various scenarios in between (13) and (14) may be admitted by choosing (14) and altering the movement coefficients such that some fraction of the stock migrates into the assigned spawning habitats.

Data Models

The basic data structure in the model is the “fleet,” which is defined here as an entity with relatively constant selection characteristics (i.e., vulnerability coefficients). In this sense a fleet can include a collection of individuals with different selection habits as long as the aggregate selection pattern does not vary much through time. Fishery-independent surveys may be regarded as fleets with negligible catch. Predators other than humans may also be treated as a “fleet” if there are some data relating to their consumption of the stocks in question.

The basic catch equation for each fleet is

$$(15) \quad C_{iasy} = \sum_h \frac{F_{iasyh}}{\xi_{iasy} Z_{iasyh}} \tilde{N}_{cah} (1 - e^{-Z_{iasyh}})$$

The term catch here refers to the total number of fish caught, including both the number landed and the number released. The fishing mortality rate coefficient F , on the other hand, refers to the fraction that are killed and explicitly includes ξ (equation 7); therefore the ξ terms cancel out in equation (15). In the present application there are four basic types of data associated with the seasonal catches of each fleet—total catch C_{iasy} , an index of abundance I_{iasy} , age composition p_{iasy} and length composition p_{ilasy} :

$$(16) \quad C_{iasy} = \sum_a C_{iasy}$$

$$(17) \quad I_{iasy} = Q_i C_{iasy} / f_{iasy}$$

$$(18) \quad p_{iasy} = C_{iasy} / C_{iasy}$$

$$(19) \quad p_{ilasy} = \sum_a p_{iasy} g\{l|a\}_{iasy}$$

where Q is a scaling coefficient used when the units of the index are not consistent with the units used for C or f , and g is a function of the growth parameters that expresses the probability that a fish from age-class a is length l . Note that the calculations are made over the entire life span of each cohort, but provision is made for cases where the data are censored (e.g., when the last category in the data are cumulative for fish older than a certain age or larger than a certain size).

One issue of concern is how best to deal with the situation where some fraction of the catch is discarded (released) and subsequently dies, i.e., how to parameterize ξ . Under the presumption that discarded fish are mostly below the size limit L ,

$$(20) \quad \xi_{iasy} = 1 - (1 - d_{ias}) G_{L|a}$$

where d is the fraction of released fish that die and $G_{L|a}$ is the probability that a captured fish will be smaller than the size limit given that it is age a . Estimates of commercial landings (harvest H) are often available, but seldom the number caught (C) or discarded (D). Assuming discarded fish are mostly below the size limit, one obtains

$$(21a) \quad H_{iasy} = (1 - G_{L|a}) C_{iasy}$$

$$(21b) \quad D_{iasy} = G_{L|a} C_{iasy}$$

The total number killed (K) is simply the sum of the harvest and the number of discarded fish that die ($d_{ias} D_{iasy}$).

Reference Points

Current law requires federal fishery management plans to include an “MSY control rule” that is comprised of two limit reference points known as the maximum fishing mortality threshold (MFMT) and the minimum stock size threshold (MSST). When the abundance of the stock drops below the MSST, provisions must be made to rebuild the stock to the level that would support the maximum sustainable yield (MSY), or some proxy thereof, within a time frame that is as short as possible commensurate with the productivity of the stock and the needs of the fishing community. A common practice of the Gulf of Mexico Fishery Management Council has been to set MFMT equal to the fishing mortality rate that will produce MSY (F_{MSY}) and to set MSST equal to a fraction ($1-M$) of the spawning potential required to sustain MSY (S_{MSY}).

The computation of reference points such as S_{MSY} is complicated by the existence of multiple fleets operating in multiple habitats on multiple stocks. For example, the MSY obtained by maximizing over all stocks simultaneously will generally be lower than the sum of values obtained when each stock is treated as though it were harvested independent of the others. Similarly, maximizing over all fleets can lead to a situation where fleets that are less efficient in terms of yield are allocated negligible effort. The approach taken here is to fix the relative allocation of effort among the ‘directed fleets’ and absolute allocation of effort among ‘bycatch fleets’ to the values desired by managers (say some particular fraction of the estimates for recent years). This tact admits the possibility that the management policies relating to the bycatch fleets may be based on considerations other than their take from the stocks in question. The MSY is then computed by rescaling the combined effort of the directed fleets so as to maximize their landings over the selected stocks and habitats. Strictly speaking, such a procedure does not produce the MSY because it is conditioned on a potentially suboptimal allocation strategy (Goodyear 1996; Powers 2005). A better term would perhaps be the maximum sustainable marginal yield (MSMY).

A difficulty with employing reference points based on MSY or MSMY concepts is that they can lead to a situation where less productive stocks are extirpated as a consequence of optimizing the exploitation of more productive stocks. A less risk-prone policy would be to adopt a strategy based on maintaining the spawning potential ratio for each stock above some predetermined minimum. The spawning potential ratio (SPR) is defined as the lifetime spawning potential of each recruit expected with a given level of fishing divided by the corresponding expectation with no fishing (Goodyear 1993). In the simplest case of one fishery, one stock, one season and one habitat it is computed as

$$(22) \quad \text{SPR} = \frac{\sum_{a=1}^A E_a e^{-\sum_{j=0}^{a-1} F v_j + M_j}}{\sum_{a=1}^A E_a e^{-\sum_{j=0}^{a-1} M_j}}$$

More generally, SPR is equivalent to ϕ when recruitment is constant. Thus, it may be calculated for various combinations of fishing effort by initializing the recursion implied by equation (14) with the same arbitrary recruitment values for all scenarios (including no fishing).

Parameter Estimation

A Bayesian approach to estimation is adopted wherein one seeks to develop a ‘posterior’ probability density for the vector of parameters Θ that is conditioned on the data, $P(\Theta | \text{data})$. By application of Bayes rule it is easy to show that

$$P(\Theta | \text{data}) \propto P(\text{data} | \Theta) P(\Theta)$$

where $P(\text{data} | \Theta)$ is the sampling density (likelihood function) and $P(\Theta)$ is the prior density (in this case the analyst’s best guess of the probability density for Θ). Estimates for Θ may be obtained by integrating the posterior (classical Bayes moment estimator)

$$(23) \quad \bar{\theta}_i = \int \theta_i P(\text{data} | \Theta) P(\Theta) d\theta_i, \quad \theta_i \in \Theta$$

or, as done here, by minimizing its negative logarithm (highest posterior density estimator, Bard 1974)

$$(24) \quad \min_{\Theta} \{ -\log_e P(\text{data} | \Theta) - \log_e P(\Theta) \}$$

Sampling densities.—Sampling densities, also known as likelihood functions, measure the disparity between the model predictions and observed data. Catch, index and effort data are assumed to be normal or lognormal distributed, e.g.,

$$(25) \quad -\log_e P(C|\Theta) = \begin{cases} 0.5 \sum_i \sum_y \sum_s \left(\frac{C_{isy}^{obs} - C_{isy}}{\sigma\{C_{isy}\}} \right)^2 - \log_e (\sigma^2\{C_{isy}\}) & \text{normal} \\ 0.5 \sum_i \sum_y \sum_s \left(\frac{\log_e(C_{isy}^{obs}/C_{isy})}{\sigma\{\log_e C_{isy}\}} \right)^2 - \log_e (\sigma^2\{\log_e C_{isy}\}) & \text{lognormal} \end{cases}$$

where the superscript *obs* distinguishes the observed data from the value predicted by the model. The variable $\sigma\{\}$ is the standard deviation of the enclosed quantity.

Data describing the age and length composition of a sample ought to be multinomially distributed provided measurement error is low and the samples were truly taken at random. In that case, the appropriate log-likelihood functions for the age and length composition of the catch are

$$(26) \quad \begin{aligned} -\log_e P(\mathbf{p}_a | \Theta) &= \sum_i \sum_y \sum_s n_{isy} \sum_a p_{iasy}^{obs} \log_e p_{iasy} \\ -\log_e P(\mathbf{p}_l | \Theta) &= \sum_i \sum_y \sum_s n_{isy} \sum_l p_{ilisy}^{obs} \log_e p_{ilisy} \end{aligned}$$

where again the superscript *obs* distinguishes the observed data from the value predicted by the model and n indicates the effective sample size input by the analyst.

Prior densities.—Prior densities are similar to sampling densities in that they measure the disparity between the model predictions of a parameter and other information known about it. The difference is that sampling densities express the probability of observing some

information (data) given the model estimates, whereas prior densities express the probability of observing the model estimates given some information (prior knowledge). When possible prior densities should be based on data, otherwise one may choose to adopt functional forms that are relatively uninformative over the plausible range of parameter values. For example, the logarithm of the natural mortality rate might be treated as uniformly distributed between -5 and 2 . The primary advantage of using uninformative priors is that the potential for introducing biases is minimized. On the other hand, if the data relating to a particular parameter are too sparse, the solution may be so uncertain as to be rendered meaningless. This observation has led some to develop prior densities based on expert opinion (e.g., Wolfson et al. 1996; Punt and Walker 1998) or analyses of other species (e.g., Liermann and Hilborn 1997; Maunder and Deriso 2003).

One parameter of special concern in the analysis of Gulf of Mexico red snapper is the steepness of the stock–recruitment relationship. Previous analyses have estimated this parameter to be close to the mathematical limit of 1.0 (Anonymous 1999), suggesting it may not be well determined. A possible alternative is to develop a prior based on a subset of the values collected by Myers et al. (1999) that corresponds to larger, highly fecund fishes with long life spans (the ‘periodic’ strategists of Rose et al. 2001). There is, of course, the potential for introducing bias when one or more of the priors are based on expert opinion or otherwise subjective information. However, the same sorts of bias can be introduced by conducting sensitivity analyses where the unknown parameters are fixed to various values selected by the analysts. It might be best to incorporate this uncertainty in a more rigorous fashion.

Covariance parameters.—It is not generally possible to obtain consistent estimates for all of the elements of the covariance matrix associated with the objective function, i.e., the correlation coefficients and variances. As is typically done in most stock assessments, each individual data record is assumed to be independent of the other data, implying the correlation coefficients are

zero. In the case of the fishery (survey) data, the variances associated with sampling variability are often estimated extraneous to the population model (e.g., during the standardization procedure). However, there may be additional variance owing to fluctuations in the distribution of the stock relative to the survey habitat (IWC 1994). To accommodate such possibilities, the variance parameters for the catches and indices of abundance may be modeled as

$$(27) \quad \sigma^2 \{C_{iy}\} = \chi^2 \{C_{iy}\} + \lambda \{C_i\} \sigma^2$$

$$\sigma^2 \{I_{iy}\} = \chi^2 \{I_{iy}\} + \lambda \{I_i\} \sigma^2$$

where the χ^2 are the annual observation variances associated with each type of data (estimated outside the model), σ^2 reflects some overall process variance (estimated within the model), and the λ are constant multipliers (usually fixed by the analyst based on a careful consideration of the inherent variability of the underlying processes). The variances corresponding to the age and length composition data are implicit functions of sample size, which is controlled on input.

The model has been implemented using the nonlinear optimization package AD model Builder (Otter Research Ltd.²), which provides facilities for estimating the mode and shape of the posterior distribution.

Application to Red Snapper

Model Structure

The model described above was applied to information on red snapper populations in the U.S. Gulf of Mexico during the years from 1872 to 2003. Five fisheries were designated for each of two regions east and west of the Mississippi River: handline, longline (after 1979), recreational (after 1945), closed-season bycatch (after 1990) and offshore shrimp trawl bycatch (after 1947). Three four-month seasons were modeled, starting in January. Spawning was assumed to occur during the second season. Each cohort was modeled for 30 years, the contribu-

tion of very old animals being assumed negligible. The initial age structure of the populations east and west of the Mississippi River were set to virgin levels as the fishery that existed prior to 1872 was by all accounts negligible. Subsequent recruitment of age-0 or age-1 fish was modeled as a Beverton–Holt function of the relative number of eggs produced (with a one-year lag in the case of age-1). Recruitment was allowed to deviate from the Beverton–Holt expectations according to equation (9) with $\rho = 0$ and $CV = 0.4$ (no deviations were estimated prior to 1930 owing to insufficient data). Separate recruitment parameters were estimated for the populations east and west of the Mississippi River. Note that the choice of starting with age 1 implies that density-dependent natural processes constitute the dominant source of mortality during the first year of life, whereas starting with age 0 implies that density-independent mortality sources (such as shrimp bycatch) dominate after settlement.

Parameter specifications.—A total of 861 parameters were estimated; 4 for the two spawner-recruit curves, 210 recruitment deviations, an overall variance parameter, and 646 parameters pertaining to the ten “fleets.” The vulnerability and catchability coefficients for each specific fleet were assumed to be relatively unchanged through time, but allowed to vary with age and among fleets (totaling 76 estimable parameters). The effect of minimum size limits was modeled explicitly as described above. While SEDAR 7 participants recognized that improvements in fishing gear since the 1870s may have resulted in changes in the vulnerability schedule, it was felt that these changes were relatively small and that the major effect would be an increase in the efficiency (effective effort) of each fleet. Accordingly, the effective effort of each fleet was allowed to vary by year essentially as a free parameter (556 estimable parameters). The scaling coefficients Q for each index of abundance were assumed to be constant through time (12 parameters). The vulnerability coefficients for the fishery independent surveys were fixed to 0 for age-1 and 1.0 for ages 2 and older. Natural mortality was fixed to 0.98 year⁻¹ for age-0, 0.6 year⁻¹ for age-1 and 0.1 year⁻¹ thereafter (SEDAR 2004). The degree of intermixing between

²Otter Research Ltd. 2001. An introduction to AD MODEL BUILDER Version 4.5. Box 2040, Sidney B.C. V8L 3S3, Canada.

the populations of red snapper east and west of the Mississippi was assumed to be negligible.

The fecundity at age (including maturity) was set to the vector derived from the age-conditioned model described by Porch (2004), normalized to have a maximum value of 1 at age 30. Thus, the spawning potential (S) of the adult population is not expressed in terms of the number of eggs produced, but in terms of the effective number of fully-productive spawners. In other words, S is the number of age 30 animals required to produce the same number of eggs as the estimated adult population.

Likelihoods and priors.—The catch, effort, and relative abundance indices were assumed to be approximately lognormal distributed. Age composition was assumed to be multinomial distributed. A lognormal prior (Nowlis 2004a) was imposed on α with a median value of 13.3 and log-scale variance of 1.28 (equivalent to a mean steepness of about 0.86). The remaining parameters were treated as free parameters constrained to lie with bounds that encompassed the range of plausible values (essentially the same as specifying uninformative priors over the feasible range).

Data employed

Landings data.—The commercial landings from 1963 to 2003 are discussed by Turner et al. (2004) and the landings prior to 1963 are discussed by Porch et al. (2004) and Porch et al. (this volume). The annual recreational harvest since 1981 is based on the NMFS Marine Recreational Fisheries Statistics Survey³ (MRFSS), Texas Parks and Wildlife Survey and NMFS headboat survey as described by Turner et al. (2004). The recreational harvest statistics used for earlier years (1946–1980) were reconstructions based on U.S. census data (Scott 2004). It is assumed that prior to 1946 the recreational take was negligible in comparison to the commercial take owing to the relative inaccessibility of the fishing grounds (powered vessels were few and expensive, making offshore trips

mostly a past time for the wealthy). The bycatch of juveniles from the offshore shrimp fishery is based on the series produced by Nichols (2004), which extends back to 1972. A time series of offshore shrimping effort, which extends back to the advent of the offshore shrimp fishery in 1948, was also used to tune the model (see Porch and Turner 2004). The catch during the closed season was derived by Turner et al. (2004).

The discards from the recreational and commercial fleets during the open season were assumed to occur predominantly due to the regulations on minimum size. They were computed on a seasonal rather than annual basis to better accommodate the rapid growth exhibited by younger red snapper. The population growth curve and coefficient of variation of length about age were fixed to the values estimated by Diaz et al. (2004).

The CV's used to weigh the landings data were fixed at 0.1 (arbitrary low value) for the commercial fleets inasmuch as they represent a census. The exceptions are for years when no census was taken, in which case the effective CV's were computed from the census estimates immediately before and after the year in question (absolute difference divided by the mean); the reasoning being that the true value likely lies somewhere between those values. The CV's for the recreational catches after 1981 came from the variance estimates produced by the MRFSS (G. Diaz, personal communication); the CV's for the catch inputs prior to 1982 were assigned arbitrary high CVs (1.0) inasmuch as they were not actually observed. The CV's for the shrimp bycatch are based on the CV's of the overall index (ages 0–2), but modified by the proportion that are not age zero (see Porch and Turner 2004). An additional process variance term was not included for the catch (cf. Equation (27); instead it is assumed that process variations in catch are adequately modeled by inter-annual deviations in recruitment and fishing mortality rates.

Indices of abundance.—Ten indices of abundance were used, 5 for each region (east or west). These include the handline CPUE series based on log books (McCarthy and Cass-Calay 2004), the MRFSS recreational indices (Cass-Calay 2004),

³Marine Recreational Fisheries Statistics Survey, National Marine Fisheries Service, Fisheries Statistics and Economics Division, 1315 East-West Highway, Silver Spring, MD 20910.

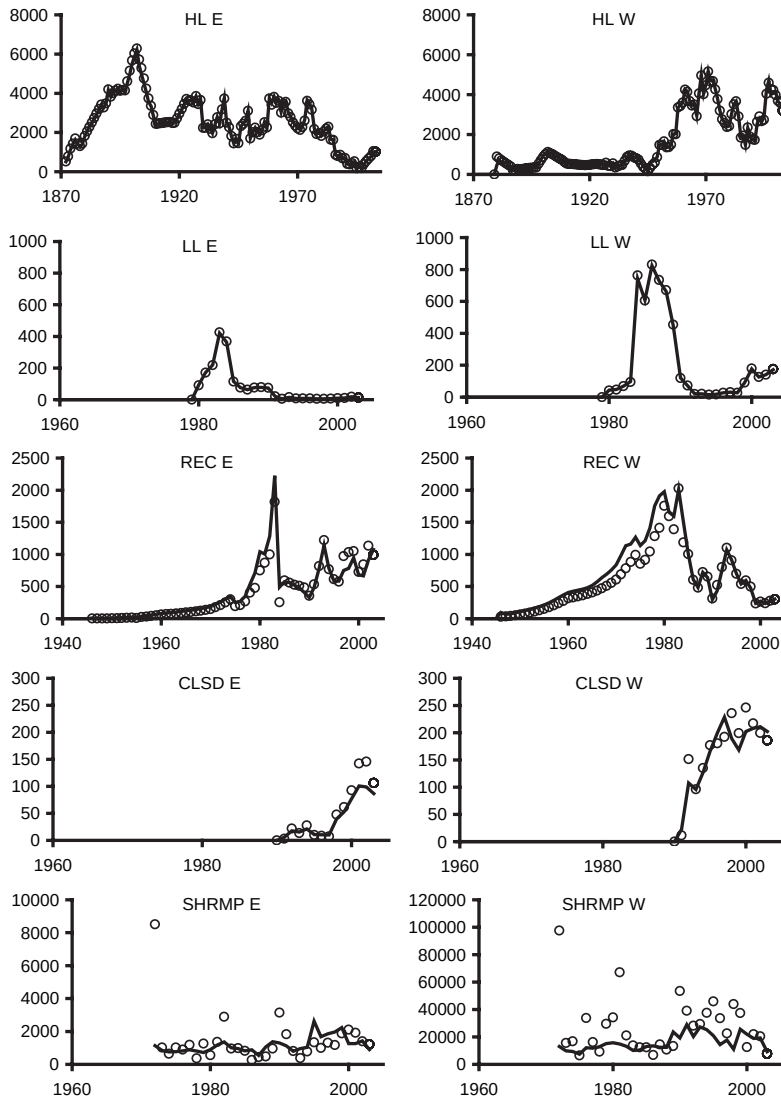


Figure 1. Age 0 model fits (lines) to the total landings (points) for each of the five “fleets” in the regions east (E) or west (W) of the Mississippi River. Figures for the handline (HL) and longline (LL) fleets are in thousands of pounds landed. Figures for the recreational fleet (REC) are in thousands of fish landed. Figures for the closed season (CLSD) and shrimp bycatch are in thousands killed. The fits obtained with the age-1 model were very similar and therefore not shown.

SEAMAP larval indices (Lyczkowski-Shultz et al. 2004), SEAMAP trawl survey (Nichols et al. 2004; Turner and Porch 2004) and video surveys (Gledhill and Ingram 2004). The handline log-book indices were modeled in this case as landings per unit effort (in pounds) rather than catch per unit effort, thereby taking into account the potential discards owing to the minimum size limit and removing the major objection to their use by the SEDAR 7 Data Workshop participants. The SEAMAP larval indices were assumed to index

spawning potential and the video surveys were assumed to index the combined abundance of ages 2 and older.

The CV's for the indices of abundance are based on the year-specific estimates that come from the GLM-based procedures used to standardize them (see the references cited above). These are regarded as representing observation variance. To this the model adds an internally-estimated process variance term, which is intended to represent random discrepancies between the

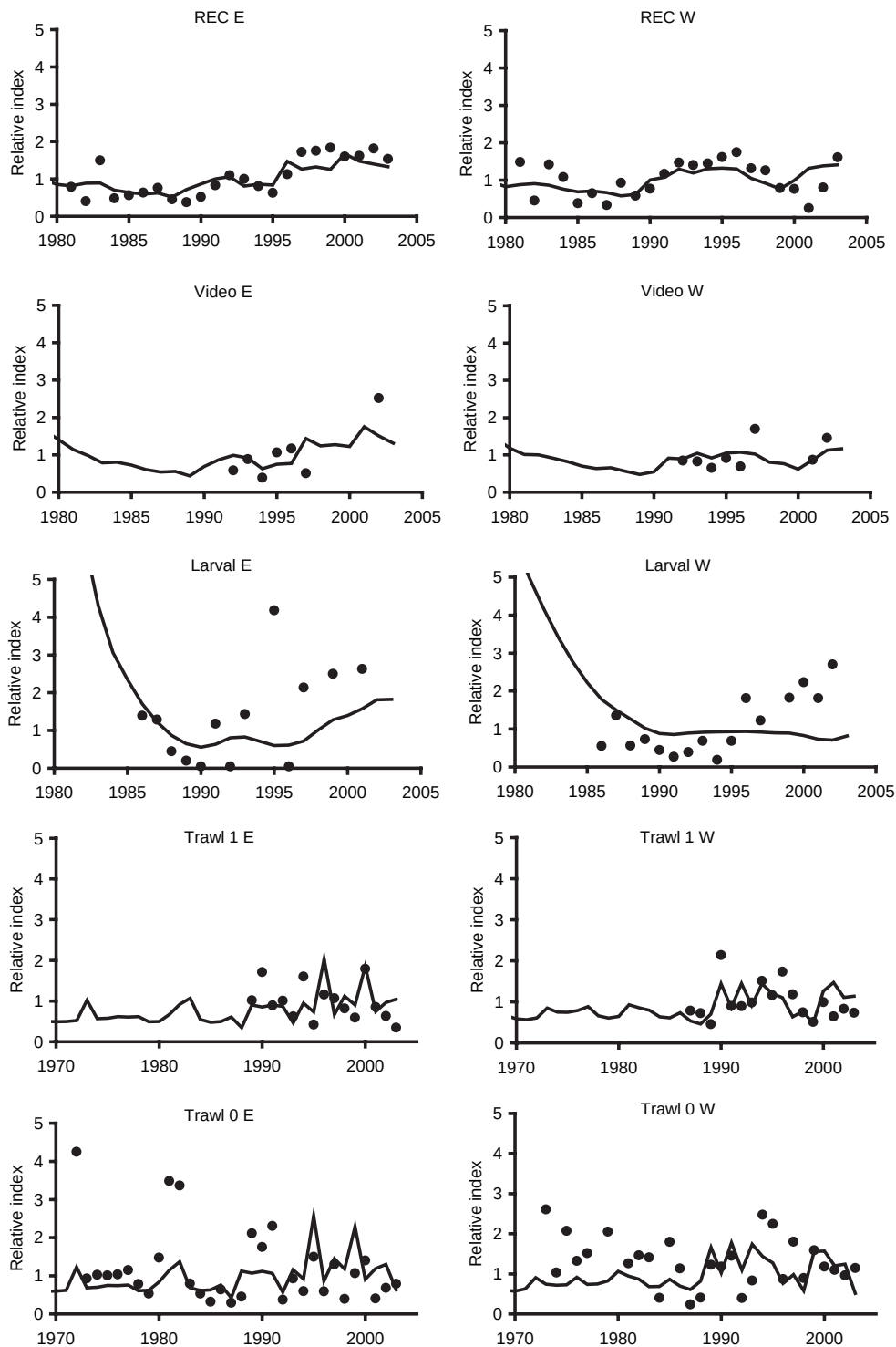


Figure 2. Age 0 model fits to indices of abundance (rescaled by the mean of the predicted values).

trends in the indices and the trends in the actual population it purports to track (see equation (27)).

Index values were rescaled to approach the magnitude expected for the population to facilitate the estimation of the catchability scaling parameters (e.g., it allows the initial guesses for q to be set to 1 and makes setting the upper and lower limits more intuitive).

Age composition.—The age composition data (and effective sample sizes) used for the commercial and recreational fisheries are described by Nowlis (2004b). Inasmuch as the model makes seasonal calculations with spawning occurring during the second season (mid-year), the data for each year were aggregated by the actual integer age in years (it is not necessary to shift the ages by 0.5 to track cohorts as VPA and ASAP must do). The age composition for the shrimp bycatch was based on model output from Nichols (2004). The age composition used for the closed season is described by Turner et al. (2004).

Results

Model fits to data

The age-0 and age-1 models both matched the total catch data quite well with the exception of the 1983 peak in the eastern recreational catch series and the high shrimp bycatch during some of the early years (Figure 1). The fit to these values appears poor because confidence in the data are low (i.e., the CV's associated with the data are high) and the model explicitly discounts the importance placed on them. The model also fit most of the indices of abundance reasonably well (Figure 2), but could not reconcile the increasing trend in the western larval index (representing spawners) with the flat or declining trends indicated by the other western indices. The model fits to the SEAMAP trawl series show a strong residual pattern where the predictions for the early years are considerably lower than the trawl values, but the predictions for the later

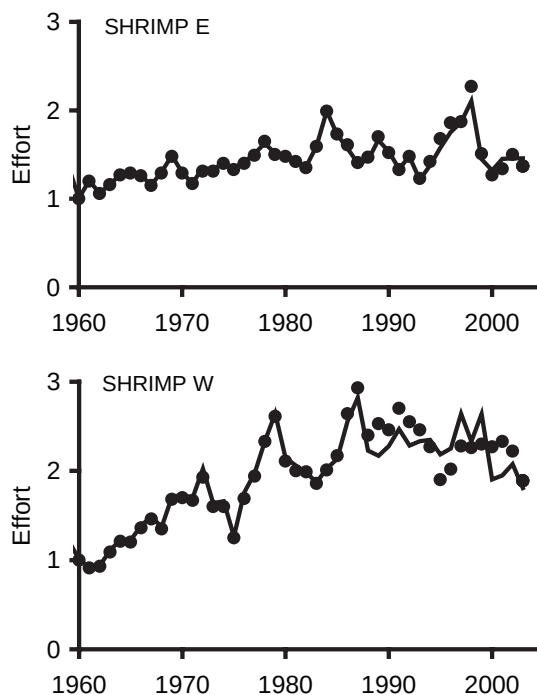


Figure 3. Age-0 model fits to the shrimp trawl effort series.

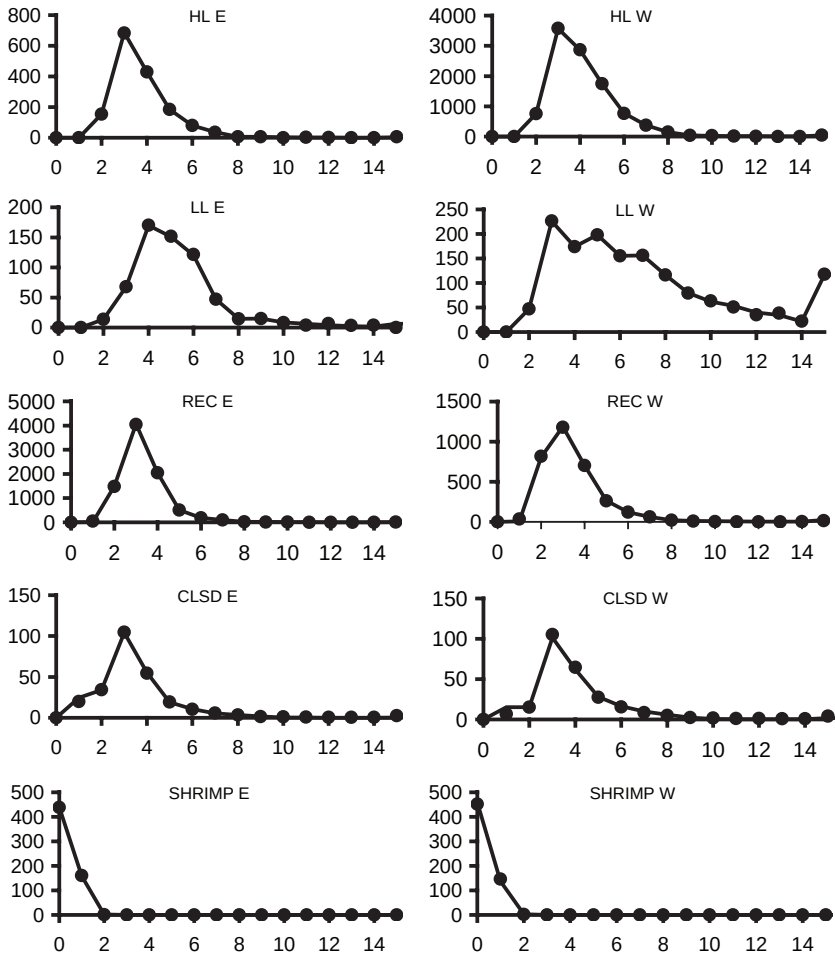


Figure 4. Age-0 model fits to the age composition data (aggregated across years).

years are considerably higher. The mismatch for the early years can be attributed to the low confidence (high CV's) associated with those data. The mismatch in more recent years reflects the influence of the bycatch data, which, in the context of relatively constant or declining effort, suggests recruitment generally has increased in recent years. The shrimp effort series were well fit (Figure 3) owing to the low CV's assigned to those data (10%).

The fits to the age composition data, aggregated over all years, appear to be quite good (Figure 4). It should be kept in mind, however, that the fits to individual years are noisier, particularly where the sample size was small.

Parameter estimates

A list of the time-invariant parameters with their highest posterior density estimates and asymptotic standard errors (computed from the inverse of the Hessian matrix) is included in Table 2. The estimates for most of these parameters appear to be reasonably precise, generally having coefficients of variation (standard error divided by the point estimate) of less than 30%. The estimates for the numerous age or time-varying parameters and associated derived quantities are shown graphically rather than tabulated (see below). The corresponding coefficients of variation were mostly less than 30%.

The estimated vulnerability and apical (fully-selected) fishing mortality rates F for the age-0 and age-1 base models are shown in Figures 5 and 6, respectively. (Recall that the vulnerabil-

ity coefficients reflect the probability of being caught rather than the probability of being landed, which also depends on the probability of being greater than the size limit.) The vulnerability of red snapper to the recreational and commercial hand line fleets follows a dome-shaped pattern with a peak at age 1 or 2 for the former and at age 5 for the latter. The vulnerability of red snapper to the commercial long line fleet follows a logistic pattern with older animals (10+) being the most vulnerable. The vulnerability patterns for the closed season “fleets” were between the hand line and longline. As expected, age 0 and age 1 fish were much more vulnerable to shrimp trawls than age 2 or older.

The estimated trends in F indicate persistent increase for all fleets. Although the recreational fishing rate in the east appears to have declined markedly in recent years, it remains at rather high levels. The highest rates were associated with the western shrimp bycatch followed by the eastern recreational and western commercial handline fisheries. Note, however, that the high shrimp bycatch rates apply only to ages 0 and 1, whereas the lower apical F 's estimated for the handline and recreational fleets apply to multiple age classes. Hence, the bycatch and directed fisheries have comparable cumulative impacts.

There does not appear to be a strong relationship between the number of recruits and

Table 2. Time-invariant parameters estimated in the models with corresponding highest posterior density (HPD) estimates, standard errors and coefficients of variation (CV). The time or age-varying parameters (vulnerability, recruitment and effort) are shown graphically to save space. The letters E and W represent the red snapper fisheries east and west of the Mississippi, respectively. The abbreviations HL, LL, REC, CLSD, SHMP represent the commercial hand line, long line, recreational, closed season and shrimp (bycatch) fleets, respectively. Trawl 1 and trawl 0 refer to SEAMAP trawl indices for age 1 and age 0 red snapper. The estimates for α were constrained by the imposed upper boundary condition of 150; therefore the CVs are near zero.

Parameter	HPD estimate	Std. error	CV (%)
R_0 E	6.58E+06	5.23E+05	8
α E	1.50E+02	2.27E-01	0
R_0 W	2.84E+07	2.36E+06	8
α W	1.50E+02	4.38E-01	0
q HL E	7.98E-02	1.70E-02	21
Q HL E	5.25E-02	1.97E-02	38
q HL W	9.07E-03	1.56E-03	17
Q HL W	5.27E-02	1.08E-02	20
q LL E	9.50E-03	3.29E-03	35
q LL W	7.60E-03	2.35E-03	31
q REC E	1.54E-02	3.78E-03	25
Q REC E	1.07E-01	1.56E-02	15
q REC E	1.33E-02	2.75E-03	21
Q REC W	3.26E-02	5.27E-03	16
q CLSD E	7.82E-03	3.34E-03	43
q CLSD W	9.34E-03	3.84E-03	41
q SHMP E	9.18E-02	1.12E-02	12
q SHMP W	3.17E-01	2.54E-02	8
Q Video E	1.20E-02	3.30E-03	27
Q Video W	5.22E-02	7.90E-03	15
Q Larval E	3.69E+00	8.88E-01	24
Q Larval W	2.77E+01	4.50E+00	16
Q Trawl 1 E	1.81E+00	2.24E-01	12
Q Trawl 1 W	9.62E-01	9.52E-02	10
Q Trawl 0 E	1.53E+00	1.33E-01	9
Q Trawl 0 W	4.66E-01	4.45E-02	10
σ^2	3.24E-01*	2.61E-02	8

*expressed as a coefficient of variation

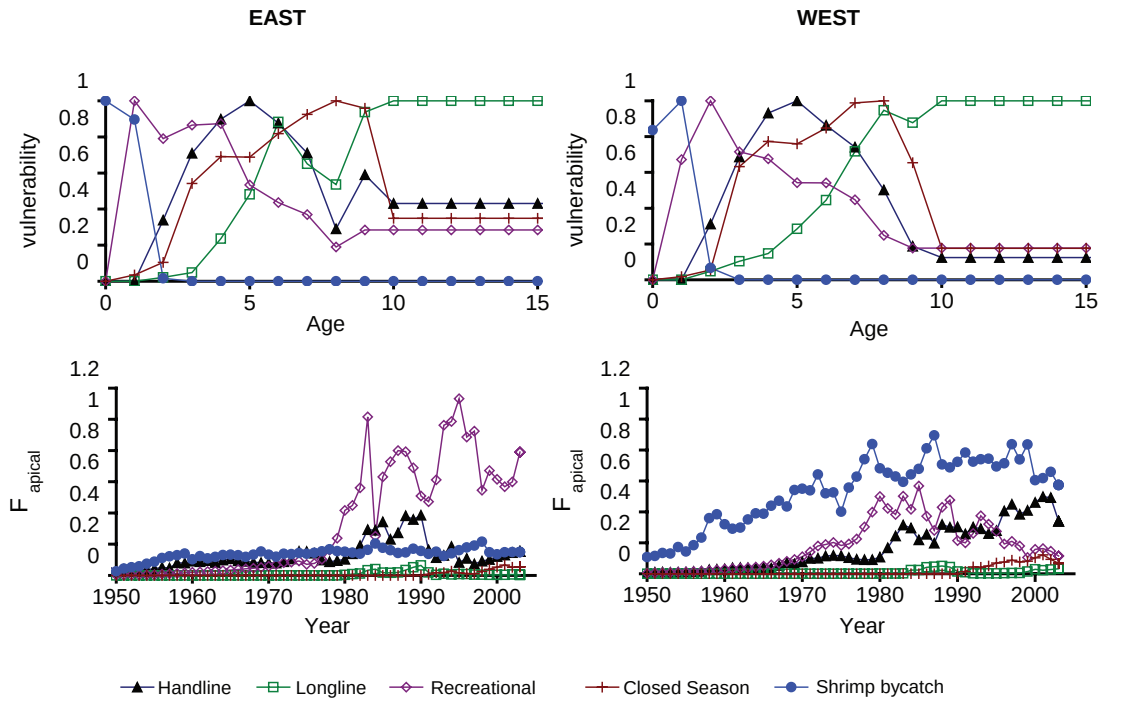


Figure 5. Age-0 model estimates of vulnerability and apical fishing mortality rate for each fleet.

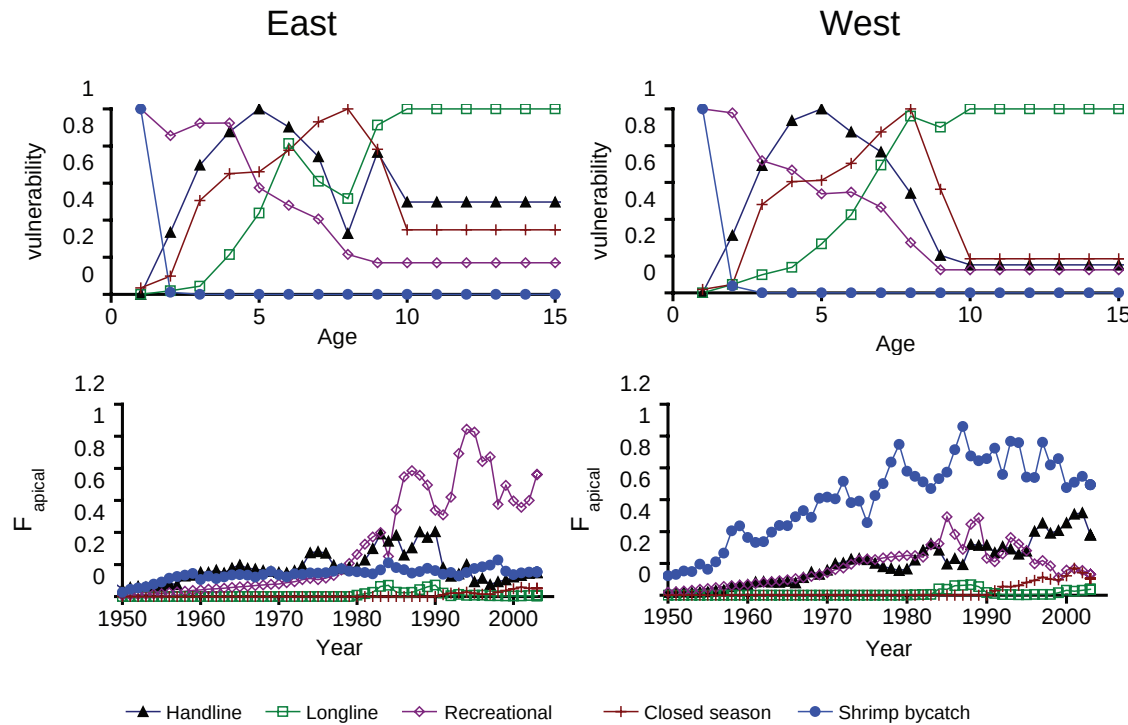


Figure 6. Age-1 model estimates of vulnerability and apical fishing mortality rate for each fleet.

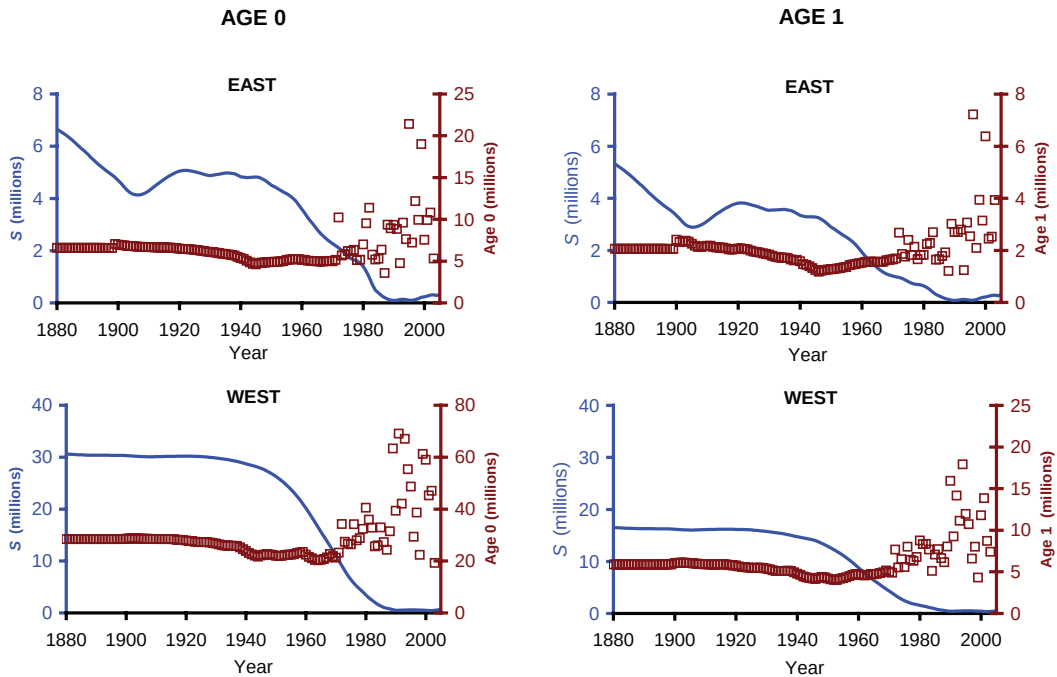


Figure 7. Age-0 and age-1 model estimates of spawning potential (lines) and corresponding number of age 0 or age 1 recruits (squares).

spawning potential (S) in the previous years (see Figure 7). In both runs the estimates of the maximum potential spawn per recruit (α) were near the limit of 151 imposed by the model, which translates to a steepness of 0.974.

Estimated population trends

Estimates of historical trends in spawning potential and recruitment are shown in Figure 7. Prior to exploitation, the western population of red snapper is estimated to have been about three times as large and three times as productive as the eastern population. The eastern population began to decline very early in the history of the fishery, whereas the western population did not decline until the late 1940s. Both populations are estimated to have been reduced to less than 10% of their unfished levels by the 1980s. The declines continued into the 1990s, when both stocks were at all time lows of less than 3% of unfished levels. In recent years the western population appears to have leveled off and the eastern population appears to have increased to a little better than 4% of pre-exploitation levels.

Discussion

The results from the age-0 and age-1 assessment models are consistent in that they both suggest the populations of red snapper in the U.S. Gulf of Mexico are severely depleted. The estimated historical trends may be somewhat biased prior to the 1950s owing to ambiguities in the fraction of the total landings that came from foreign waters (see Porch et al. 2007, this volume), but nevertheless agree well with published anecdotes. The decline in the east is estimated to have begun shortly after the fishery began and continued into the early 1900s, which is consistent with the observation that some of the largest catches on record occurred during that time. The population appears to have rebounded somewhat between the 1920s and early 1940s owing to the shift to the Campeche Banks and reduced effort during events such as the Great Depression and World War II. Shortly after the war, however, the population began to plummet; reminiscent of Camber's (1955) claim that most of the snapper banks off Florida were considered impoverished. Such a decline is not surprising given the widespread adoption of postwar innovations

such as diesel engines, depth finders, reels, and wire lines. The continued decline through more recent years is also not unexpected inasmuch as the snapper fleet nearly doubled in size during the boat building boom of the late 1950s and early 1960s (Moe 1963; Carpenter 1965). Vessel fishing power has also continued to improve owing to various gear modifications, electronic fish-finders and navigational aids such as loran-c and GPS. Even so, there is some evidence of an increase in recent years, possibly due to regulations implemented during the 1990s to rebuild the stock (see Hood et al. 2007, this volume).

The decline in the west is estimated to have begun during the late 1940s and to have continued through the 1990s. Although productive fishing grounds had been discovered off Texas as early as the 1880s (e.g., the “Galveston Lumps”), most of the region was poorly charted and believed to be relatively unproductive insofar as red snapper were concerned. The turning point came during the 1940s with innovations such as diesel engines, depth recorders, reels and wire lines, which made it much easier to find and exploit snapper grounds in deeper waters. One of the most productive new areas was known as the “Western” grounds, a huge area off Texas and Louisiana spanning 10 mi on either side of the 100 fathom (180 m) line. Its discovery apparently led to a shift from the prevailing paradigm that the center of abundance of red snapper in U.S. waters was east of the Mississippi (c.f. Jordan and Evermann 1923) to a view that the center of abundance lay west of the Mississippi. As Camber (1955) remarks, “the largest populations seem to concentrate in the Gulf of Mexico, especially off the Yucatan Peninsula and the Texas and Louisiana coast.” This view is supported by the present assessment, which estimates that the potential (unfished) abundance of the western population is about three times that of the eastern population.

Despite the apparent enormity of the western red snapper resource, the increased effort associated with the boat building boom and technological advances discussed earlier seems to have taken its toll. In fact, the models indicate that the western stock may have been depleted even more rapidly than the smaller eastern stock. It should be pointed out that none of the indices of

abundance used to tune the model extend prior to the 1970s, therefore the slope of the declines is probably not very well determined. On the other hand, Bradley and Bryan (1975) reported that Texas handliners landed about 1,000 lbs (454 kg) per day prior to 1965, but felt fortunate to catch 500 lbs (227 kg) per day during the early 1970s. The implication then is that the available stock had decreased by somewhat more than 50%, which is very close to the model estimates of a 52% decline from 1960–1964 to 1970–1973.

One of the more remarkable features of the present assessment is the increase in both the magnitude and inter-annual variability of the recruitment estimates over the last several decades. The increase in variability is commensurate with the beginning of the SEAMAP trawl survey, which provides information on variations in year-class strength that allows the model to admit substantial departures from the static spawner-recruit relationship. What is more interesting is the fact that the model estimates of recruitment since 1980 are mostly well above the estimates for an unfished population despite the estimated decrease in spawning potential. The structure of the model does not require the recruitment deviations from the estimated spawner-recruit relationship to sum to zero, which to some extent decouples the “pre-historic” trends dictated by the spawner-recruit relationship from the more recent trends estimated when more data are available. Nevertheless, the estimated steepness is near 1.0 despite the much smaller value imposed in the prior, suggesting that the spawner-recruit relationship is reasonably well-determined. Thus, a possible interpretation of the peculiar pattern in recruitment is that the red snapper stocks in the Gulf of Mexico have become more productive over the last two decades.

A number of possible mechanisms have been postulated to account for the elevated recruitment estimates for recent years. They include improved oceanographic conditions for larvae, larval input from the Campeche Banks, postsettlement density-dependent mortality effects, reduced predator abundance (possibly due to shrimp trawling), increased habitat for juveniles (oil rigs and artificial reefs) and unde-

tected increases in spawning potential (cryptic adult biomass). At present, the data available are insufficient to discriminate between these hypotheses. Regardless, it is likely that the recruitment estimates for the 1980–2003 period are better-determined than the earlier values as this is the period when most of the data are available. Accordingly, it is probably more reasonable to assume that future recruitment will follow recent levels than to assume they will follow the estimated spawner-recruit relationship (which presumably reflects the earlier history of the fishery). For this reason the participants of the SEDAR 7 review workshop recommended that forecasts and reference points be based on an alternative spawner-recruit relationship with the R_0 values for each stock set equal to the average of the recruitment estimates for 1980 to 1903 (Cordue 2005).

The determination as to whether or not a stock is over-fished depends of course on the reference point the stock is measured against. Several different reference points are routinely computed for the Gulf of Mexico red snapper fishery, but none of these have been formalized in the Gulf of Mexico Fishery Management Council's fishery management plan. As a result, the implications of changes in the stock assessment methodology or data are often confused with the implications of changing the reference points. To date, the reference points that have received the most attention are based on the MSMY or SPR achieved under two types of effort allocation schedules, referred to hereafter as the "equal proportion" and "percent shrimp reduction" schedules. The equal proportion scenario assumes the effective effort of all fleets, both directed and undirected, can be scaled

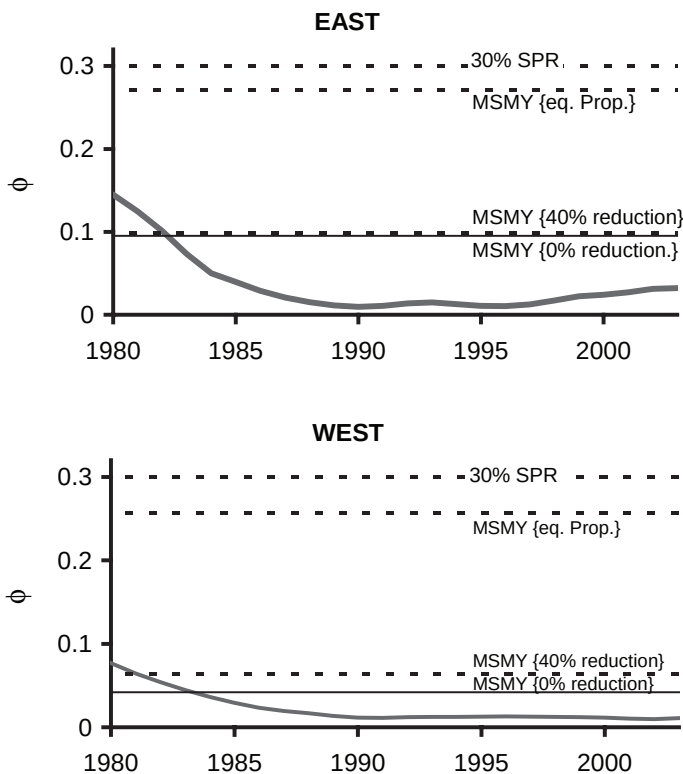


Figure 8. Age-0 model estimates of relative spawning potential ϕ (heavy solid line) compared to four reference points: ϕ_{MSMY} given current shrimp bycatch (fine horizontal line), ϕ_{MSMY} given a 40% reduction in shrimp bycatch rates, ϕ_{MSMY} when all bycatch and directed fleets reduced by the same proportion, and $\phi_{30\% \text{ SPR}}$.

Table 3. Estimates of the sustainable annual landings (millions of pounds) under four harvest strategies

Strategy	Model	East	West	Total
MSMY 0% reduction	Age 0	4.5	6.8	11.3
MSMY 0% reduction	Age 1	4.4	6.3	10.7
MSMY 40% reduction	Age 0	5.1	11.5	16.5
MSMY 40% reduction	Age 1	4.7	9.3	14.1
MSMY equal proportion	Age 0	6.4	19.0	25.4
MSMY equal proportion	Age 1	5.2	12.9	18.1
SPR30% equal proportion	Age 0	6.4	18.9	25.3
SPR30% equal proportion	Age 1	5.1	12.8	18.0

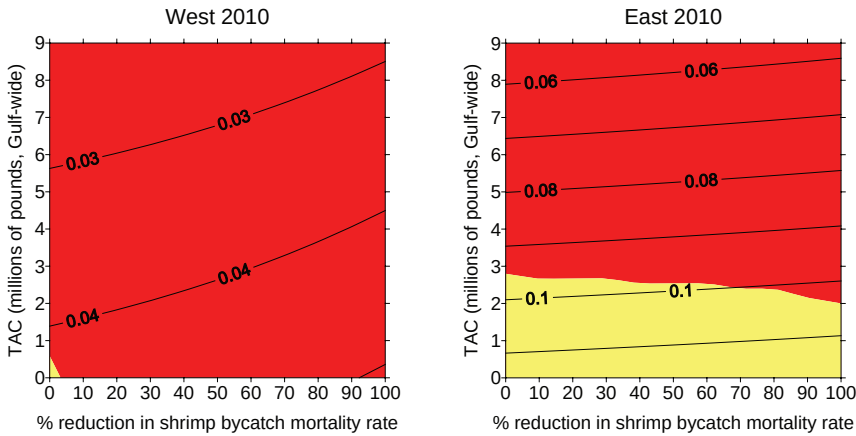
down by the same proportion. This has sometimes been referred to as the “linked” or “policy neutral” scenario because all of the fleets that catch red snapper are affected to the same degree. The percent shrimp reduction schedule recognizes that the management of offshore shrimp fisheries is not completely driven by concerns about red snapper bycatch and therefore that any prescribed reduction in offshore shrimp effort will be somewhat independent of the reductions in the directed fisheries. It also assumes that the fraction of the red snapper population killed during the closed season will be similar to recent levels (here the 2001–2003 average) when the directed fisheries are scaled down. Otherwise, the relative allocation of effort among the directed fleets has been assumed to remain at recent levels.

The impact of different reference points on the perception of stock status is illustrated for the age-0 model in Figure 8. Note that the metric of comparison, $\phi = S/S_0$, is equivalent to SPR for steepness values near 1.0, as is presently the case. The spawning potential of both the East and West stocks is estimated to have been well below each of the reference points considered, therefore one must conclude that the two stocks are currently overfished. The degree of imperilment and ability of the stock to recover, however, depends on which reference point is considered most reasonable. The MSMY policies predicated on only a limited reduction in offshore shrimp bycatch are clearly more risky than a policy based on 30% SPR because they permit the stock to be fished down to within

only a few percent of the unfished levels. When shrimping and closed season bycatch rates are assumed to continue at the 2001–2003 average, for example, ϕ_{MSMY} amounts to 4.2% and 9.5% for the west and east, respectively. When shrimping is assumed to be reduced by 40%, the ϕ_{MSMY} values are only a little higher at 6.4% and 9.8% for the west and east. In both cases the ϕ values are well below the generic recommendation of 30% adopted by the Gulf of Mexico Fishery Management Council. The values of ϕ_{MSMY} were much closer to 30% under the equal proportion reduction scenario, amounting to 26% and 27% of unfished levels for the west and east, respectively. In that case the MSMY was achieved when the F of all fleets was reduced by 74%, including shrimp bycatch and closed-season discards. (Recall, the percent reduction scenarios assume that closed-season discarding will continue at the 2001–2003 rate.)

One disturbing feature of previous assessments of Gulf of Mexico red snapper has been the tendency to estimate MSMY values that are much greater than the historical landings and yet predict the stock to be overfished. This paradoxical behavior is not uncommon where the time series of data are short because the model has little basis for determining the historical development of the fishery. Indeed, it was partly for this reason that SEDAR 7 participants recommended extending the historical catch series back to the inception of the fishery in 1872. The models shown here, which use the extended time series, do in fact produce more plausible MSMY estimates (Table 3). Even so,

Age 0 model



Age 1 model

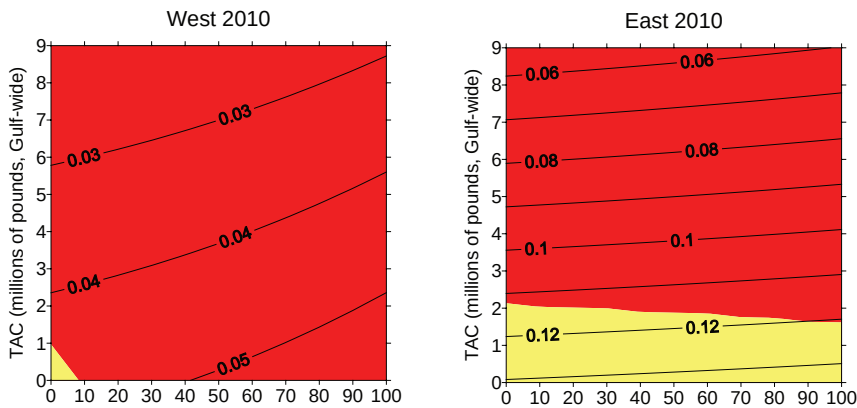


Figure 9. Isopleths of relative spawning potential in the year 2010, $\phi_{2010} = S_{2010}/S_0$, for the age-0 (top) and age-1 (bottom) models. The horizontal axis refers to the projected percent reduction in shrimp bycatch mortality rate and the vertical axis refers to the projected Gulf-wide TAC. The shading represents different levels of spawning potential relative to MSMY levels (S_{2010}/S_{MSMY}), where MSMY is conditioned on the indicated reduction in shrimp bycatch mortality. Red represents $S_{2010}/S_{MSMY} < 1$ and yellow represents $1 < S_{2010}/S_{MSMY} < 4$.

the estimates of MSMY for the equal proportion scenario are substantially greater than even the highest recorded landings. This is primarily a result of the use of the higher recruitment estimates for recent years (1980–2003) as indicative of the present potential of the stock, which of course implies that MSMY and S_{MSMY} levels are greater now than they were in the past. In other words, the models suggest that (1) the stocks were less productive in the past and therefore more easily overfished; and (2) the stocks are more productive now, but their potential has not been fully realized because they had already been depleted.

One possible consequence of defining a ref-

erence point based on recent increases in productivity is that the stock can appear overfished without overfishing having ever occurred. Consider, for example, a fully-exploited stock ($F = F_{MSMY}$ and $S = S_{MSMY}$) that has experienced a 50% increase in recruitment in recent years owing to a change in the environment. This would translate roughly into a 50% increase in the value of S_{MSMY} , but it would take some time for the additional recruits to contribute to the spawning potential of the stock. Accordingly, if the new higher level of S_{MSMY} were adopted as the reference point, then the stock would initially be reclassified as overfished regardless of the actions of the fishery. In practice the change from

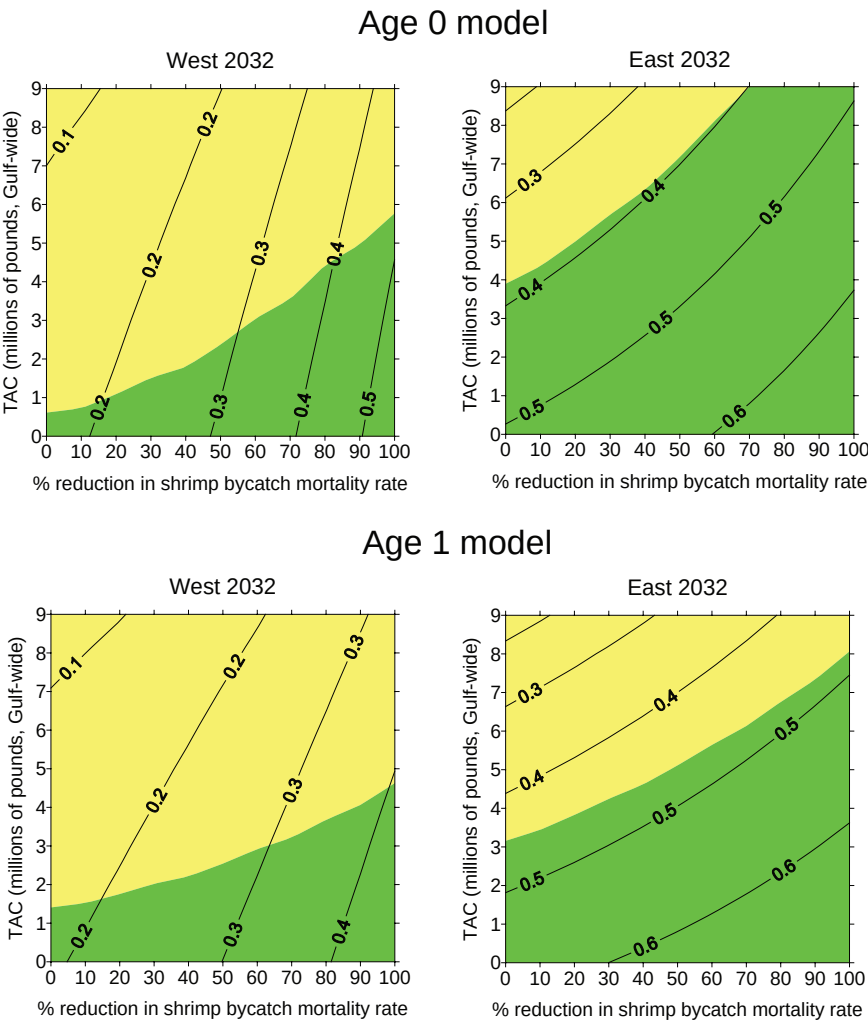


Figure 10. Isopleths of relative spawning potential in the year 2032, $\phi_{2032} = S_{2032}/S_0$, for the age-0 (top) and age-1 (bottom) models. The horizontal axis refers to the projected percent reduction in shrimp bycatch mortality rate and the vertical axis refers to the projected Gulf-wide TAC. The shading represents different levels of spawning potential relative to MSMY levels (S_{2032}/S_{MSMY}), where MSMY is conditioned on the indicated reduction in shrimp bycatch mortality. Yellow represents $1 < S_{2032}/S_{MSMY} < 4$ and green represents $S_{2032}/S_{MSMY} > 4$.

one productivity level to another may be rather gradual and difficult to establish. Nevertheless, the issue raises the prospect of developing reference points that distinguish the effect of overfishing from the transitional effect of changes in productivity.

In the case of red snapper, the transition to a higher productivity regime appears to have begun prior to 1980. If past catch levels had been sustainable before the new regime, then similar levels of catch over the last two decades should have resulted in a substantial increase in spawn-

ing potential. Instead, the eastern population has exhibited only modest gains and the western population appears not to have increased at all. This implies that, despite their increased reproductive potential, the two stocks had been so depleted in the past that they will remain unable to recover without further restrictions on fishing. The question then arises as to what level of catch and bycatch mortality might permit the recovery of the stock to its potential under the new higher recruitment scenario within a reasonable time frame.

Figure 9 presents isopleths of S/S_{MSMY} and S/S_0 (ϕ) generated from short-term projections of the western and eastern stocks to the year 2010 under various levels of TAC (Gulf-wide total allowed landings, assuming the relative levels of effort in the east and west remain constant) and percent reductions in shrimp bycatch mortality rate (F_{shrimp}). Current (2001–2003) conditions are assumed to prevail until 2007, when the indicated TACs and bycatch reductions take effect. The value of S_{MSMY} is conditioned on the reduction in F_{shrimp} indicated on the horizontal axis. Hence, the graph should be interpreted as an indication of what might happen if managers based the MSMY definition on the actual reduction in shrimp bycatch rates (implying that managers can either control or accurately forecast future shrimp bycatch). The projections suggest the current TAC of 9 million lb (4,100 mt) is sustainable, but unlikely to foster a recovery from the current low levels ($\phi_{2003} = 2\%$ in the west and 4% in the east). Substantial short-term gains in spawning potential might be achieved with TACs under 4 million pounds, particularly in the east. Reductions in F_{shrimp} are projected to have little impact because the short term recovery in spawning potential is driven largely by the above average recruitments estimated to have occurred during the late 1990s.

The situation is more optimistic with a longer recovery time (Figure 10). The spawning potential of both stocks is projected to exceed S_{MSMY} by 2032 even at current levels of TAC and F_{shrimp} . However, it should be kept in mind that the SPR and ϕ values associated with MSMY are rather low unless the percent reduction of F_{shrimp} is large. If a more conservative reference point such as 30% SPR were to be adopted, then either the TAC or F_{shrimp} must be reduced considerably. In the case of the east, the estimates of F_{shrimp} are substantially less than the mortality rates for the directed fisheries, therefore a recovery to $S_{30\% \text{ SPR}}$ ($\phi > 30\%$) is predicted for TACs between 6 and 9 mp and less than a 40% reduction in F_{shrimp} . In the case of the west, where shrimp bycatch rates are much higher, a recovery to $S_{30\%}$ appears unlikely with less than a 50% reduction in F_{shrimp} , even in the absence of a directed fishery. Conversely, a recovery to $S_{30\%}$ is possible with the current TAC if F_{shrimp} were reduced by at least 75%.

The age-0 model assigns greater importance to the shrimp bycatch fishery than does the age-1 model (which ignores the bycatch of age 0 animals). Hence, for a given TAC, a smaller percent reduction in F_{shrimp} is estimated to be required to bring about recovery. For example, projections based on the age-0 model suggest that a 7 mp TAC combined with a 70% reduction in shrimp bycatch rate would likely allow the spawning potential of both stocks to recover the level associated with a 30% SPR. Projections with the age-1 model, however, indicate that the shrimp bycatch must be reduced by more than 80%. Not surprisingly, there has been some debate over which of the two model formulations presented is most plausible. Both formulations relate the number of recruits to past spawning potential by use of the Beverton–Holt function, which implies a belief that the survival from the egg phase to the age of recruitment is dictated primarily by density-dependent processes such as competition for resources. Thus the crux of the debate is the matter of timing. The age-0 model implies that any density-dependent effects on survival occur primarily during the planktonic phase or shortly after settlement. The age-1 model, on the other hand, implies that density-dependent effects dominate throughout the first year of life and that density-independent sources of mortality such as shrimp bycatch can be ignored.

The SEDAR 7 review workshop participants (Cordue 2005) selected the age-0 model as the most plausible of the formulations presented, stating that it “was not aware of any other assessment where the possibility that density-dependent compensatory processes occurring simultaneously with density-independent mortality from fishing (either discards or retained catch) was considered justification for treating the mortality from fishing as insignificant.” It is important to remember, however, that the recruitment estimates of both models were allowed to deviate substantially from the predictions of the estimated Beverton–Holt functions. As a result, the results from the two models are not as disparate as one might imagine. Nevertheless, there is some evidence that density-dependent processes may be important during the first several years of life and it would be useful to model postrecruitment density dependence more directly.

Recent studies of red snapper life history characteristics, CPUE trends and otolith microchemistry suggest that there is a rather strong demarcation between the populations living east and west of the Mississippi river (Cowan et al. 2002). Nevertheless, tag-recapture studies have shown that red snapper occasionally move substantial distances (e.g., Patterson et al. 2001) and otolith microconstituent analyses indicate that some fish move from one side of the river to the other (Cowan et al. 2002). Hence, it may be prudent to extend the analyses presented here to allow for some degree of intermixing. The parameters for the movement model described above could be estimated from age-composition samples identified to stock by use of equations (18) and (19) with additional subscripts to reference stock, but a correction factor would be needed to account for the effect of misclassification errors.

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Appendix

Re-parameterized spawner-recruit relationships

The number of young fish recruiting to a population (R) is often related to spawning potential (S) using one of two functional forms:

$$(A. 1) \quad R = \begin{cases} aSe^{-bS} & \text{Ricker} \\ \frac{aS}{b+S} & \text{Beverton and Holt} \end{cases}$$

The parameter a is the slope of the curve at the origin and the parameter b controls the degree of density dependence. Notice that the domain of both functions extends from zero to infinity, whereas in practice there must be some limitation on S and R even in the absence of fishing owing to environmental constraints (call them S_0 and R_0 , respectively). This being so, we obtain

$$(A. 2) \quad a \frac{S_0}{R_0} = \begin{cases} e^{bS_0} & \text{Ricker} \\ 1 + S_0/b & \text{Beverton and Holt} \end{cases}$$

The ratio S_0/R_0 represents the maximum expected lifetime fecundity of each recruit and a represents the survival of recruits in the absence of density dependence. Accordingly, the product $\alpha = aS_0/R_0$ may be interpreted as maximum possible number of recruits produced by each spawner over its lifetime (Myers et al. 1999).

The dimensionless character of α makes it useful for interspecies comparisons, or for borrowing values from species with similar life history strategies. Solving for b in terms of α one obtains

$$(A. 3) \quad b = \begin{cases} \log_e \alpha / S_0 & \text{Ricker} \\ S_0 / (\alpha - 1) & \text{Beverton and Holt} \end{cases}$$

Substituting (A. 3) into (A. 1) gives

$$(A. 4) \quad R = \begin{cases} aS\alpha^{-S/S_0} & \text{Ricker} \\ \frac{aS_0}{1 + (\alpha - 1)S/S_0} & \text{Beverton and Holt} \end{cases}$$

and, since $a = \alpha R_0/S_0$,

$$(A. 5) \quad R = \begin{cases} R_0 \frac{S}{S_0} \alpha^{1-S/S_0} & \text{Ricker} \\ R_0 \frac{\alpha S/S_0}{1 + (\alpha - 1)S/S_0} & \text{Beverton and Holt} \end{cases}$$

Defining $\phi = S/S_0$ gives equation (9).

Note that when spawning extends over multiple seasons in the model, but the same spawner-recruit function is used for each season, then R_0 and S_0 should be interpreted as the virgin levels associated with a particular reference season. In that case, R_0 and S_0 will not necessarily be greater than the virgin values associated with other seasons.